

Technical Specification and Systems Engineering Blueprint for the Static Holographic Boundary Theory Synthetic Gravitational Lensing Telescope (SGLT)

Abstract—The Static Holographic Boundary Theory Synthetic Gravitational Lensing Telescope (SGLT) is specified as a two-craft deep-space observatory that replaces a conventional primary mirror with a controlled synthetic metric perturbation. Boundary-state congestion generates an effective ghost seed of $M_{\text{seed}} = 10^{-6} M_{\odot}$, establishing a focal baseline $f_0 = 169.30$ m for impact parameter $r_0 = 1.000$ m. Metric projection is performed by a cryogenic 8×8 InP/InGaAs heterojunction-bipolar-transistor transducer array with niobium routing, sapphire acoustic transport, and helium-4 termination. A 1,800-module lattice-assisted nuclear reaction power plant supplies 913.176 kW net electrical output at a verified thermoelectric-generator efficiency of 33.804%; 4.6550 MW of waste heat is rejected by 688.52 m² of effective radiator area at 600 K. The plant provides fourteen-module redundancy above the 1,786-module debt threshold, and the formation carries 5370 kg dry mass, 1190 kg propellant, and 6560 kg wet mass with numerical margins and qualification reserves. Dual-wavelength heterodyne metrology attains 0.144 pm/ $\sqrt{\text{Hz}}$ single-axis range noise, 0.772 nm three-sigma formation tracking, and 11.38 nrad differential-wavefront-sensing attitude knowledge. Deterministic control is provided by the bare-metal shbt-os microkernel using a statically allocated 2,112-byte SRAM frame, a 1.412 ns AVX-512 SIMD shunt decision, 9.24 ns post-quench recovery, and 0.561 μs POSIX shared-memory latency. The synthesized aperture achieves 0.06289-arcsec angular resolution and Strehl ratio $S = 0.999999984$. At Sun–Earth L2, closed-loop formation control combines EMI–BF₄ electrospray actuation with electro-optic phase correction while maintaining eigenvector detuning below 10^{-12} . The resulting architecture closes the relativistic, optical, electrical, thermal, computational, navigation, and fault-containment interfaces required for qualification against an eighteen-gate integrated verification baseline.

Index Terms—artificial gravitational lensing, boundary conformal field theory, formation flight, heterojunction bipolar transistor, InP/InGaAs, lattice-assisted nuclear reaction, metric engineering, Sun–Earth L2, synthetic aperture.

I. MISSION ARCHITECTURE AND SYSTEM OVERVIEW

A. Mission-Level Physical Baseline

The SGLT shall synthesize an effective point-mass perturbation by commanding the boundary-state excess ΔN above the admissible local-state limit. The verified coupling law and

reference state are

$$\begin{aligned} M_{\text{seed}} &= \alpha_{\text{seed}} \Delta N, \\ \Delta N &= N_{\text{local}} - N_{\text{limit}}, \\ \alpha_{\text{seed}} &= 1.3258 \times 10^{-51} M_{\odot} \text{ bit}^{-1}, \\ \Delta N_0 &= 7.5426 \times 10^{44} \text{ bit}, \\ M_{\text{seed},0} &= \alpha_{\text{seed}} \Delta N_0 = 9.99998 \times 10^{-7} M_{\odot} \simeq 10^{-6} M_{\odot}. \end{aligned} \quad (1)$$

With the CODATA gravitational constant, defined speed of light, and nominal solar mass,

$$\begin{aligned} G &= 6.67430(15) \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}, \\ c &= 299\,792\,458 \text{ m s}^{-1}, \\ M_{\odot} &= 1.98847(7) \times 10^{30} \text{ kg}, \end{aligned} \quad (2)$$

the effective Schwarzschild radius and thin-lens focal distance are

$$\begin{aligned} R_{s,0} &= \frac{2GM_{\text{seed},0}}{c^2} = 2.95334 \times 10^{-3} \text{ m}, \\ f_0 &= \frac{r_0^2}{2R_{s,0}}, \\ r_0 &= 1.000 \text{ m}, \\ f_0 &= 169.30 \text{ m}. \end{aligned} \quad (3)$$

The mission-level stability condition is imposed directly on the holographic transport eigenvector:

$$|\delta\mu| \leq \delta\mu_{\text{floor}} = 1.0 \times 10^{-12}. \quad (4)$$

The boundary-condition data set declares a bit-congestion radius and an operational exclusion radius as

$$\begin{aligned} R_{\text{congestion}} &= 2.954 \times 10^{15} \text{ m} = 1.9746 \times 10^4 \text{ AU}, \\ R_{\text{excl}} &= 19.7 \text{ AU} = 2.9471 \times 10^{12} \text{ m}, \\ 1 \text{ AU} &= 1.495978707 \times 10^{11} \text{ m}. \end{aligned} \quad (5)$$

Equation (5) prevents the dimensionally inconsistent identification of 2.954×10^{15} m with 19.7 AU. The two values belong to distinct boundary and bulk domains and are related by the calibrated holographic scale factor developed in Section II-B; they are not alternate unit representations of one physical radius. Mission assurance shall apply the declared congestion radius to boundary-capacity and numerical-stability analyses and the 19.7-AU radius to commanded inter-seed separation. Sun–Earth L2 selection mitigates externally driven detuning through low differential acceleration and continuous formation observability; it does not waive either radius constraint.

B. Two-Craft Formation and L2 Operations

The flight segment comprises a Lens Craft and a Sensor/Focal Craft. The Lens Craft carries the metric-field generator, 8×8 cryogenic transducer array, LANR electrical plant, helium-4 acoustic sink, radiator wings, deterministic control computer, and emergency current-diversion network. The Sensor/Focal Craft carries the photon-counting focal-plane assembly, spectrometers, wavefront sensor, optical crosslinks, and cold-gas/microelectric formation actuators. Their commanded relative state is

$$\begin{aligned} \mathbf{r}_{\text{SF/LC}} &= f_0 \hat{\mathbf{k}} + \delta \mathbf{r}, \\ f_0 &= 169.30 \text{ m}, \\ \|\delta \mathbf{r}\|_{3\sigma} &= 0.772 \text{ nm} < 1.00 \text{ nm}, \\ \|\delta \mathbf{v}\|_{3\sigma} &\leq 5.00 \times 10^{-10} \text{ m s}^{-1}, \end{aligned} \quad (6)$$

where $\hat{\mathbf{k}}$ is the instantaneous optical-axis unit vector. The verified heterodyne and differential-wavefront-sensing observables are

$$\begin{aligned} S_r^{1/2} &= 0.144 \text{ pm}/\sqrt{\text{Hz}}, \\ \|\delta \mathbf{r}\|_{3\sigma} &= 0.772 \text{ nm}, \\ \sigma_{\theta, \text{DWS}} &= 11.38 \text{ nrad}, \\ S_r^{1/2} &< 0.170 \text{ pm}/\sqrt{\text{Hz}}, \quad \sigma_{\theta, \text{DWS}} < 15.00 \text{ nrad}. \end{aligned} \quad (7)$$

The baseline Sun–Earth L2 halo orbit shall satisfy an annual station-keeping allocation below 2.5 m s^{-1} , compared with an atmospheric-drag-dominated requirement exceeding 150 m s^{-1} in low Earth orbit (LEO):

$$\begin{aligned} \Delta V_{\text{L2}} &< 2.5 \text{ m s}^{-1} \text{ yr}^{-1}, \\ \Delta V_{\text{LEO}} &> 150 \text{ m s}^{-1} \text{ yr}^{-1}. \end{aligned} \quad (8)$$

Table I defines the deployment trade using the formation dry mass of 5.370 Mg established in Table II, a ten-year service interval, a 30% maneuver reserve, and electric propulsion with a specific impulse of 1600 s. The site-trade reserve is computed independently of the installed 1190-kg propellant load. The reserved propellant follows

$$\begin{aligned} m_{p,10} &= m_d \left[\exp \left(\frac{1.30 n \Delta V_{\text{yr}}}{g_0 I_{\text{sp}}} \right) - 1 \right], \\ m_d &= 5370 \text{ kg}, \quad n = 10 \text{ yr}, \\ g_0 &= 9.80665 \text{ m s}^{-2}, \quad I_{\text{sp}} = 1600 \text{ s}. \end{aligned} \quad (9)$$

C. Integrated Mass, Power, and Thermal Closure

The master resource allocation in Table II establishes the dry and wet masses, installed propellant, continuous bus power, maneuver capability, and telecommunications interfaces for both flight elements. All operational sites retain the same science hardware, power plant, and thermal rejection architecture.

The allocated maneuver capabilities are vehicle-level quantities and are not additive at formation level.

At the reference seed mass, the bus allocation and the redundant 1,800-module LANR generation ledger close as

$$\begin{aligned} P_{\text{debt}} &= \left(\frac{M_{\text{seed},0}}{M_{\odot}} \right) (906 \times 10^9 \text{ W}) = 906.00 \text{ kW}, \\ N_{\text{min}} &= \left\lceil \frac{906000 \text{ W}}{507.3195 \text{ W module}^{-1}} \right\rceil = 1786, \\ N_{\text{LANR}} &= 1800 = N_{\text{min}} + 14, \\ P_{\text{net}} &= 913.176 \text{ kW}, \\ \Delta P_{\text{gen}} &= P_{\text{net}} - P_{\text{debt}} = 7.176 \text{ kW}, \\ \eta_{\text{TEG}} &= 33.804\%, \\ P_{\text{th}} &= 5.5682 \text{ MW}, \\ P_{\text{rej}} &= P_{\text{th}} - P_{\text{net}} = 4.6550 \text{ MW}. \end{aligned} \quad (10)$$

The 906.0726 kW Lens-Craft bus entry in Table II is the allocated continuous load, whereas P_{net} is the installed generating capability. For emissivity $\epsilon = 0.92000$, radiator temperature $T_r = 600 \text{ K}$, background temperature $T_b = 2.7 \text{ K}$, and Stefan–Boltzmann constant $\sigma = 5.670374419(86) \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$, the effective radiator area is

$$\begin{aligned} A_r &= \frac{P_{\text{rej}}}{\epsilon \sigma (T_r^4 - T_b^4)} \\ &= 688.52 \text{ m}^2, \\ \delta A_r|_{\delta T_r = \pm 2 \text{ K}} &= \mp 9.18 \text{ m}^2. \end{aligned} \quad (11)$$

The structured gate record and the flat simulation export shall be isomorphic under the numerical crosswalk below. JSON array indices are zero based; each CSV field contains the scalar mapped by the corresponding `gates[i].measured` member. All eighteen gates constitute the integrated acceptance set.

D. Transducer, Lithography, and Deterministic-Control Interfaces

The primary metric projector is a 64-channel InP/InGaAs HBT array on a $50.0 \text{ } \mu\text{m}$ pitch. Table III is the controlling GDSII mask-interface definition; all coordinates shall be snapped to a 1.0-pm database grid.

The bare-metal `shbt-os` controller shall execute without dynamic allocation from a 2112-byte UnifiedStinespringFrame, partitioned into a 640-byte active workspace and a 1472-byte environmental-state ledger. The SHBT-MMIO-1 interface occupies the physical base address `0x70000000` and implements the register map in Table IV.

The verified AVX-512 monitor has an issue-to-MMIO decision latency of 1.412 ns. A violation of Eq. (4) shall assert `SHUNT_TRIG`, reduce all HBT gate commands below the 3.800-V turn-on threshold, and initiate the recovery sequencer according to

$$\begin{aligned} t_{\text{SIMD}} &= 1.412 \text{ ns}, \\ t_{\text{shunt}} &\leq 2.50 \text{ ns}, \\ t_{\text{hold}} &= 110.00 \text{ ns}, \\ t_{\text{sequence}} &= 9.24 \text{ ns}, \\ t_{\text{recover}} &= 119.24 \text{ ns} < 120.00 \text{ ns}, \\ V_{\text{gate},i}(t \geq t_{\text{shunt}}) &< 3.800 \text{ V}, \quad i \in \{0, 1, \dots, 63\}. \end{aligned} \quad (12)$$

TABLE I
DEPLOYMENT-SITE MASS, POWER, THERMAL, AND STABILITY TRADE

Deployment site	Disturbance environment	ΔV allocation (m/s/yr)	10-yr propellant (kg)	Wet mass (Mg)	Net / rejected power (MW)	Rigidity disposition
LEO, 550 km	Drag, eclipse cycling, Earth gravity gradient, atomic oxygen	> 150	> 711	> 6.081	0.9132/4.6550	Noncompliant; continuous drag and gravity-gradient forcing exceed the eigenvector disturbance allocation.
Earth–Moon L5	Earth–Moon gradient, solar pressure, weak long-period libration	35	158	5.528	0.9132/4.6550	Marginal; periodic differential acceleration consumes the formation and rigidity margins.
Sun–Earth L2 halo	Solar pressure, weak halo-orbit instability, no atmospheric drag	< 2.5	< 11.2	< 5.382	0.9132/4.6550	Baseline; full compliance with Eq. (4) under closed-loop metrology.
Heliocentric, 1.1 AU	Solar pressure and gravity; increased communications latency	< 0.5	< 2.3	< 5.373	0.9132/4.6550	Compliant; lowest maneuver load, with reduced command and ranging availability.

TABLE II
MASTER SPACECRAFT RESOURCE ALLOCATION

Element	Dry mass (kg)	Propellant (kg)	Wet mass (kg)	Propellant system	Bus power (kW)	Δv (m/s)	Telecommunications
Lens Craft	4 250.0	850.0	5 100.0	EMI–BF ₄ electro-spray / N ₂	906.0726	52.370	25.0 Mbps S/Ka-band
Sensor Craft	1 120.0	340.0	1 460.0	N ₂ and microelectric formation propulsion	4.80	62.370	12.0 Gbps laser optical downlink
Total system	5 370.0	1 190.0	6 560.0	Dual-craft allocation	910.8726	Vehicle-specific	S/Ka command plus optical science downlink

The resulting baseline closes the mission geometry, formation-control, mass, electrical, thermal, lithographic, software, and hardware-protection interfaces required to enter preliminary design at Sun–Earth L2.

II. RELATIVISTIC GRAVITATIONAL LENSING AND MULTI-SEED WAVEFRONT OPTICS

A. Linearized ADM Metric and Mass–Congestion State Equation

The lensing field is represented in a 3 + 1 Arnowitt–Deser–Misner (ADM) foliation with lapse α_{ADM} , shift β^i , and spatial metric γ_{ij} . The spacetime interval is

$$\begin{aligned} ds^2 &= -\alpha_{\text{ADM}}^2 c^2 dt^2 + \gamma_{ij} dX^i dX^j, \\ dX^i &= dx^i + \beta^i c dt, \\ \alpha_{\text{ADM}}^{(0)} &= 1, \quad \beta^{i(0)} = 0, \quad \gamma_{ij}^{(0)} = \delta_{ij}. \end{aligned} \quad (13)$$

For a static, weak perturbation in isotropic coordinates and metric signature $(-, +, +, +)$, the synthesized Newtonian po-

tential and first-order metric are

$$\begin{aligned} \Phi(r) &= -\frac{GM_{\text{seed}}}{r}, \quad r = \sqrt{x^2 + y^2 + z^2}, \\ g_{\mu\nu} &= \eta_{\mu\nu} + h_{\mu\nu} + \mathcal{O}(h^2), \\ \eta_{\mu\nu} &= \text{diag}(-1, 1, 1, 1), \\ h_{00} &= -\frac{2\Phi}{c^2} = \frac{R_s}{r}, \\ h_{0i} &= 0, \\ h_{ij} &= -\frac{2\Phi}{c^2} \delta_{ij} = \frac{R_s}{r} \delta_{ij}. \end{aligned} \quad (14)$$

The mass–congestion constitutive equation maps the localized state excess directly to the effective lens mass:

$$\begin{aligned} M_{\text{seed}} &= \alpha_{\text{seed}} (N_{\text{local}} - N_{\text{limit}}) = \alpha_{\text{seed}} \Delta N, \\ \alpha_{\text{seed}} &= 1.3258 \times 10^{-51} M_{\odot} \text{ bit}^{-1}, \\ \Delta N &= \frac{M_{\text{seed}}}{\alpha_{\text{seed}}}. \end{aligned} \quad (15)$$

The associated gravitational length scale is obtained by equating the weak-field coefficient to the Schwarzschild perturbation:

$$\begin{aligned} R_s &= \frac{2GM_{\text{seed}}}{c^2} = \frac{2G\alpha_{\text{seed}}\Delta N}{c^2}, \\ \hat{\alpha}_{\text{GR}}(r_0) &= \frac{4GM_{\text{seed}}}{c^2 r_0} = \frac{2R_s}{r_0}. \end{aligned} \quad (16)$$

EIGHTEEN-GATE JSON/CSV SIMULATION CROSSWALK

Gate	Domain	JSON scalar	CSV field	Verified value	Acceptance bound
01	Metric physics	gates[0].measured	adm_det_error	0.0	$ \det(g) + 1 \leq 1.00 \times 10^{-12}$
02	Metric physics	gates[1].measured	chi_shbt	1.000000	$ \chi_{\text{SHBT}} - 1 \leq 10^{-6}$
03	Metric physics	gates[2].measured	register_bits	512 bit	≥ 512 bit
04	GNC/optics	gates[3].measured	heterodyne_beat_hz	80 000 002 Hz	80 MHz $\pm$ 10 Hz
05	GNC/optics	gates[4].measured	range_asd_pmrtHz	0.144 pm/ $\sqrt{\text{Hz}}$	≤ 0.170 pm/ $\sqrt{\text{Hz}}$
06	GNC/optics	gates[5].measured	tracking_3sigma_nm	0.772 nm	≤ 1.000 nm
07	GNC/optics	gates[6].measured	dws_nrad	11.38 nrad	≤ 15.00 nrad
08	Transducer FEA	gates[7].measured	sapphire_mrayl	44.1782 MRayl	44.178 $\pm$ 0.010 MRayl
09	Transducer FEA	gates[8].measured	matching_layer_nm	6.3951 nm	6.395 $\pm$ 0.005 nm
10	LANR power	gates[9].measured	lanr_net_kw	913.176 kW	≥ 906.000 kW
11	LANR power	gates[10].measured	teg_efficiency_pct	33.804%	$\geq 33.800\%$
12	LANR power	gates[11].measured	radiator_area_m2	688.52 m ²	≥ 688.52 m ²
13	HIL kernel	gates[12].measured	sram_frame_bytes	2 112 byte	exact 2 112 byte
14	HIL kernel	gates[13].measured	simd_shunt_ns	1.412 ns	< 2.500 ns
15	HIL kernel	gates[14].measured	recovery_ns	9.24 ns	≤ 120.00 ns
16	HIL kernel	gates[15].measured	posix_shm_us	0.561 μ s	< 1.000 μ s
17	Optical reconstruction	gates[16].measured	resolution_arcsec	0.06289''	$\leq 0.06290''$
18	Optical reconstruction	gates[17].measured	strehl	0.999999984	≥ 0.999999980

TABLE III
CONTROLLING GDSII LAYER DEFINITION

Layer	Purpose code	Material	Thickness / geometry
10	SUBSTRATE_INP	SI InP (100)	350.0 μ m wafer
20	AIRBRIDGE_SPAN	Au/InP	1.5 μ m $\times$ 5.0 μ m
25	MET_NB_TRACE	Nb	300.0 nm; $T_c = 9.20$ K

Under the paraxial approximation, a ray incident parallel to the optical axis intersects that axis at

$$f(r_0) = \frac{r_0}{\hat{\alpha}_{\text{GR}}(r_0)} = \frac{r_0^2}{2R_s} = \frac{c^2 r_0^2}{4GM_{\text{seed}}}, \quad (17)$$

$$r_0 = 1.0000 \text{ m.}$$

The complete commanded operating range is given in Table V. Entropy-debt power follows the prescribed linear mass law.

$$P_{\text{debt}} = \left(\frac{M_{\text{seed}}}{M_{\odot}} \right) (906 \times 10^9 \text{ W}). \quad (18)$$

The tabulated bit counts are rounded to seven significant digits. Their relative quantization and constant-propagation bounds are

$$\left| \frac{\delta(\Delta N)}{\Delta N} \right| \leq 5.0 \times 10^{-7},$$

$$\left| \frac{\delta R_s}{R_s} \right| \leq \sqrt{\left(\frac{\delta G}{G} \right)^2 + \left(\frac{\delta M_{\text{seed}}}{M_{\text{seed}}} \right)^2}, \quad (19)$$

$$\left| \frac{\delta f}{f} \right| \leq 2 \left| \frac{\delta r_0}{r_0} \right| + \left| \frac{\delta R_s}{R_s} \right|.$$

B. Theoretical Closure of Bit-Congestion vs. Exclusion Radius

The declared asymptotic bit-congestion radius and the bulk operational exclusion radius are coordinates in different do-

main. Their common-unit values and direct dimensionless map are

$$R_{\text{congestion,decl}} = 2.954 \times 10^{15} \text{ m} = 0.09573 \text{ pc},$$

$$R_{\text{excl,op}} = 19.7 \text{ AU} = 2.9471 \times 10^{12} \text{ m},$$

$$\chi_{\text{SHBT}} \equiv \frac{R_{\text{congestion,decl}}}{R_{\text{excl,op}}} = 1.002341 \times 10^3, \quad (20)$$

$$R_{\text{congestion,decl}} = \chi_{\text{SHBT}} R_{\text{excl,op}}.$$

Thus, χ_{SHBT} is a boundary-to-bulk coordinate calibration, not an astronomical-unit conversion. In particular, $2.954 \times 10^{15} \text{ m}$ is 0.09573 pc; a value of 0.312 pc would instead be approximately $9.63 \times 10^{15} \text{ m}$.

The proposed quantum-gravity combination is dimensionless, but its numerical value at the reference seed mass does not by itself produce the scale in Eq. (20). Direct substitution gives

$$\chi_{\text{QG}} \equiv \left(\frac{\hbar c}{GM_{\text{seed},0}^2} \right)^{-1/4}, \quad (21)$$

$$M_{\text{seed},0} = 10^{-6} M_{\odot},$$

$$\chi_{\text{QG}} = 9.5584 \times 10^{15}.$$

Consequently, a dimensionless SHBT matching coefficient is required for a mathematically closed constitutive map:

$$\chi_{\text{SHBT}} = C_{\text{SHBT}} \left(\frac{\hbar c}{GM_{\text{seed},0}^2} \right)^{-1/4}, \quad (22)$$

$$C_{\text{SHBT}} \equiv \frac{\chi_{\text{SHBT}}}{\chi_{\text{QG}}} = 1.04865 \times 10^{-13}.$$

Equation (22) separates the postulated mass-dependent quantum factor from the empirically declared boundary-to-bulk calibration. Setting $C_{\text{SHBT}} = 1$ would contradict the two stated radii by more than twelve orders of magnitude.

For K simultaneously active seeds, the metric is decomposed into isolated weak-field responses and the non-additive

TABLE IV
SHBT-MMIO-1 FLIGHT-CONTROL REGISTER MAP

Address	Symbol	Access	Width	Required function
0x70000000	CR0_CTRL	R/W	32 bit	Enable, operating-mode select, phase-loop arm, and synchronous soft reset.
0x70000004	SR0_STAT	R	32 bit	Quench-detect, phase-lock, thermal-alarm, metrology-valid, and shunt-latched status.
0x70000008	DET_MU_HI	R	32 bit	Upper word of the signed Q1.62 eigenvector-detuning telemetry value.
0x7000000C	DET_MU_LO	R	32 bit	Lower word of the signed Q1.62 eigenvector-detuning telemetry value.
0x70000010	BIT_OVF_N	R/W	32 bit	Upper command word of the 64-bit state-overflow mantissa.
0x70000014	BIT_OVF_L	R/W	32 bit	Lower command word of the 64-bit state-overflow mantissa.
0x70000018	RF_PHASE_V	R/W	32 bit	Sixteen-bit channel address and sixteen-bit DAC code spanning 3.800–7.400 V.
0x7000001C	SHUNT_TRIG	W	32 bit	Nonzero write directly fires the solid-state emitter-current diversion network.
0x70000020	REC_SEQ_PC	R/W	32 bit	Recovery entry-point command, microsequence program counter, and terminal-state code.
0x70000040	INT_TENS_0	R/W	512 bit	Base of the 64-byte interference-tensor command aperture.

TABLE V
SYNTHETIC-SEED LENSING AND ENTROPY-DEBT OPERATING REGIMES FOR $r_0 = 1.0000$ m

$M_{\text{seed}}/M_{\odot}$	State overflow ΔN (bit)	R_s (m)	Focal distance f	P_{debt}
1.0×10^{-7}	7.542616×10^{43}	2.953339×10^{-4}	1.692999 km	90.60 kW
1.0×10^{-6}	7.542616×10^{44}	2.953339×10^{-3}	169.2999 m	906.00 kW
1.0×10^{-5}	7.542616×10^{45}	2.953339×10^{-2}	16.92999 m	9.060 MW
1.0×10^{-4}	7.542616×10^{46}	2.953339×10^{-1}	1.692999 m	90.60 MW
1.0×10^{-3}	7.542616×10^{47}	2.953339	0.1692999 m	906.0 MW

overlap correction:

$$g_{\mu\nu}(x) = \eta_{\mu\nu} + \sum_{i=1}^K h_{\mu\nu}^{(i)}(x) + \delta g_{\mu\nu}^{\text{overlap}}(x),$$

$$\delta g_{\mu\nu}^{\text{overlap}}(x) = \sum_{i < j} \mathcal{C}_{\mu\nu}^{\alpha\beta\gamma\delta} h_{\alpha\beta}^{(i)}(x) h_{\gamma\delta}^{(j)}(x) W\left(\frac{d_{ij}}{R_{\text{excl,op}}}\right),$$

$$W(q) = 0 \quad \text{for } q \geq 1, \quad (23)$$

where $\mathcal{C}$ is the configuration-controlled SHBT coupling tensor, d_{ij} is inter-seed separation, and W is the qualified overlap window. Let K_0 denote the nominal holographic transport operator, $\delta K[\delta g^{\text{overlap}}]$ its overlap-induced perturbation, and V the eigenvector matrix of K_0 . First-order eigenvalue perturbation is bounded by

$$|\delta\mu_a| \leq \kappa_2(V) \|\delta K[\delta g^{\text{overlap}}]\|_2,$$

$$\max_a |\delta\mu_a| \leq \delta\mu_{\text{floor}} = 1.0 \times 10^{-12}. \quad (24)$$

The flight computer shall therefore inhibit a second seed unless both $d_{ij} \geq R_{\text{excl,op}}$ and the bound in Eq. (24) is satisfied. An exclusion-radius violation, invalid scale-map coefficient, inconsistent seed ephemeris, or $|\delta\mu| > 10^{-12}$ asserts the hardware shunt predicate in Eq. (98); the deterministic sequence in Eq. (99) then either restores an ECC-clean, geometry-valid state or leaves the metric field inhibited.

C. Multi-Seed Superposition and the 512-Bit Interference Tensor

A single monopolar seed has an inverse focal power proportional to r_0^{-2} and therefore produces a longitudinal caustic

rather than a planar focal surface. The controlled multi-seed metric is

$$g_{\mu\nu} = \eta_{\mu\nu} + \sum_{i=1}^K h_{\mu\nu}^{(i)} + I_{\mu\nu}, \quad (25)$$

where $I_{\mu\nu}$ is the actively commanded interference tensor. The uncorrected radial focal power is

$$\mathcal{P}_r(r_0) = \frac{1}{f(r_0)} = \frac{2R_s}{r_0^2},$$

$$\frac{d\mathcal{P}_r}{dr_0} = -\frac{4R_s}{r_0^3}. \quad (26)$$

For reference aperture radius $r_{\text{ref}} = 1.0000$ m, the diagonal compensation command is defined by

$$I_{00}(r_0) = -\frac{R_s}{r_0} + \frac{2R_s r_0}{r_{\text{ref}}^2},$$

$$I_{11}(r_0) = I_{22}(r_0) = I_{33}(r_0) = I_{00}(r_0), \quad (27)$$

$$I_{\mu\nu}(r_0) = 0, \quad \mu \neq \nu.$$

At the $10^{-6} M_{\odot}$ operating point and $r_0 = r_{\text{ref}}$, all four diagonal coefficients reduce to one 512-bit binary value. The rug/MPFR representation, evaluated with round-to-nearest and ties-to-even, is specified exactly by the sign–mantissa–exponent tuple

$$\mathcal{I}_{512} = (+1) 0xC18CD00F59F01543276459773BBF9420$$

$$079F061062F313216FF7064EAF5C3D2$$

$$9C7248835CEF0BB3C726C4C9A51009A3$$

$$089BDAF38750E9B1523880972C33DD05 \times 2^{-520}. \quad (28)$$

Its exact 155-significant-digit decimal rendering is equivalently decomposed into base-10 integer blocks as

$$\begin{aligned} \mathcal{I}_{512} = & 2953339382066878558096523387881 \times 10^{-33} \\ & + 6292562558714204710746977471712 \times 10^{-64} \\ & + 0232515199820836970136357290438 \times 10^{-95} \\ & + 1962215104297226199323620401181 \times 10^{-126} \\ & + 3775828345593770442070814599714 \times 10^{-157}. \end{aligned} \quad (29)$$

Consequently, the baseline tensor coefficients and off-diagonal constraints are

$$\begin{aligned} I_{00}(1 \text{ m}) &= \mathcal{I}_{512}, \\ I_{11}(1 \text{ m}) &= \mathcal{I}_{512}, \\ I_{22}(1 \text{ m}) &= \mathcal{I}_{512}, \\ I_{33}(1 \text{ m}) &= \mathcal{I}_{512}, \\ \max_{\mu \neq \nu} |I_{\mu\nu}| &< 10^{-122}. \end{aligned} \quad (30)$$

The 512-bit significand provides a binary rounding unit and a decimal precision envelope of

$$\begin{aligned} u_{512} &= 2^{-513} = 3.729170365600103 \times 10^{-155}, \\ p_{10} &= 512 \log_{10} 2 = 154.1273578 \text{ digits}, \\ \epsilon_{\text{holo}} &= 1.0 \times 10^{-122}, \\ u_{512} &< \epsilon_{\text{holo}}. \end{aligned} \quad (31)$$

The compensated metric changes the ray deflection from the monopolar inverse-radius law to a linear aperture-radius law:

$$\begin{aligned} \hat{\alpha}_{\text{GR}}(r_0) &= \frac{2R_s}{r_0}, \\ \hat{\alpha}_{\text{eff}}(r_0) &= \frac{2R_s r_0}{r_{\text{ref}}^2}, \\ \mathcal{P}_{\text{eff}}(r_0) &= \frac{\hat{\alpha}_{\text{eff}}(r_0)}{r_0} = \frac{2R_s}{r_{\text{ref}}^2}, \\ \frac{d\mathcal{P}_{\text{eff}}}{dr_0} &= 0. \end{aligned} \quad (32)$$

Every paraxial ray within the commanded aperture therefore intersects a common plane:

$$\begin{aligned} f_{\text{planar}} &= \frac{r_0}{\hat{\alpha}_{\text{eff}}(r_0)} = \frac{r_{\text{ref}}^2}{2R_s}, \\ f_{\text{planar}} &= 169.2998789898903 \text{ m} \approx 169.30 \text{ m}. \end{aligned} \quad (33)$$

D. Electronic Beam Steering and Voltage Mapping

The 8×8 InP/InGaAs HBT array is indexed by $m, n \in \{0, \dots, 7\}$ and has pitch $d_e = 50.0 \text{ } \mu\text{m}$. Coordinates relative to the electrical phase center are

$$\begin{aligned} x_m &= \left(m - \frac{7}{2}\right) d_e, \\ y_n &= \left(n - \frac{7}{2}\right) d_e, \\ d_e &= 50.0 \times 10^{-6} \text{ m}. \end{aligned} \quad (34)$$

For commanded wavefront tilts θ_x and θ_y , wavelength λ , and tensor-correction phase $\phi_{I,mn}$, each complex element command is

$$\begin{aligned} k_0 &= \frac{2\pi}{\lambda}, \\ \vartheta_{m,n} &= -k_0 (x_m \sin \theta_x + y_n \sin \theta_y) + \phi_{I,mn}, \\ a_{m,n} &= A_{m,n} \exp(i\vartheta_{m,n}), \\ \Theta_{\text{tilt}} &= \sqrt{\theta_x^2 + \theta_y^2}. \end{aligned} \quad (35)$$

The synthesized aperture factor and electronically translated beam maximum follow

$$\begin{aligned} \mathcal{A}(k_x, k_y) &= \sum_{m=0}^7 \sum_{n=0}^7 a_{m,n} \exp[i(k_x x_m + k_y y_n)], \\ k_{x,\text{max}} &= k_0 \sin \theta_x, \\ k_{y,\text{max}} &= k_0 \sin \theta_y. \end{aligned} \quad (36)$$

The phase command is wrapped into one 2π interval and mapped linearly between the HBT gate/base turn-on voltage and collector-drain saturation voltage:

$$\begin{aligned} \bar{\vartheta}_{m,n} &= \text{mod}(\vartheta_{m,n}, 2\pi), \\ V_{\text{on}} &= 3.800 \text{ V}, \\ V_{\text{sat}} &= 7.400 \text{ V}, \\ V_{m,n} &= V_{\text{on}} + \frac{V_{\text{sat}} - V_{\text{on}}}{2\pi} \bar{\vartheta}_{m,n}, \\ V_{m,n} &= 3.800 \text{ V} + (0.5729577951 \text{ V rad}^{-1}) \bar{\vartheta}_{m,n}. \end{aligned} \quad (37)$$

The 16-bit phase DAC code and the corresponding RF_PHASE_V register write at address 0x70000018 are

$$\begin{aligned} D_{m,n} &= \text{round} \left[65535 \frac{V_{m,n} - 3.800 \text{ V}}{3.600 \text{ V}} \right], \\ q &= 8m + n, \\ W_{m,n} &= (q \ll 16) \mid D_{m,n}, \\ 0 \leq q \leq 63, \quad 0 \leq D_{m,n} \leq 65535, \\ \Delta V_{\text{LSB}} &= \frac{3.600 \text{ V}}{65535} = 54.9325 \text{ } \mu\text{V}, \\ \Delta \vartheta_{\text{LSB}} &= \frac{2\pi}{65535} = 9.58753 \times 10^{-5} \text{ rad}. \end{aligned} \quad (38)$$

Closed-loop interpolation shall reduce the raw DAC quantization residual to the phase-stability allocation by dithering adjacent codes over the 72.0-GHz timing backplane:

$$\begin{aligned} f_{\text{clk}} &= 72.0 \text{ GHz}, \\ R_{\text{route}} &= 40.0 \text{ Gb s}^{-1} \text{ channel}^{-1}, \\ |\delta \vartheta_{m,n}| &\leq 5.05 \times 10^{-5} \text{ rad}. \end{aligned} \quad (39)$$

This command chain supplies non-mechanical two-axis retargeting while preserving the constant focal power of Eq. (32) and the 169.30-m planar detector interface of Eq. (33).

III. METRIC TRANSDUCER EMITTER ARRAY AND ACOUSTIC TAMPING ASSEMBLY

A. Heterojunction Bipolar Transducer Array Architecture

The metric transducer is a monolithic 8×8 InP/InGaAs heterojunction bipolar transistor (HBT) lattice. The channel

coordinates, physical aperture, master timing, and deterministic state-routing capacity are

$$\begin{aligned}
 (m, n) &\in \{0, \dots, 7\}^2, & N_{\text{ch}} &= 64, \\
 d_e &= 50.0 \text{ } \mu\text{m}, \\
 L_{\text{ctr}} &= 7d_e = 350.0 \text{ } \mu\text{m}, \\
 f_{\text{master}} &= 72.0 \text{ GHz}, \\
 R_{\text{bus}} &= 40.0 \text{ Gb s}^{-1} \text{ channel}^{-1}, \\
 R_{\text{aggregate}} &= 64R_{\text{bus}} = 2.560 \text{ Tb s}^{-1}.
 \end{aligned} \tag{40}$$

The 72.0-GHz clock period and its relationship to the 19.82-MHz airbridge mode are

$$\begin{aligned}
 T_{\text{master}} &= \frac{1}{f_{\text{master}}} = 13.8889 \text{ ps}, \\
 \frac{f_{\text{master}}}{f_{\text{bridge}}} &= \frac{72.0 \times 10^9}{19.82 \times 10^6} = 3.63269 \times 10^3.
 \end{aligned} \tag{41}$$

All mask vertices shall be quantized to the GDSII database grid

$$\Delta_{\text{GDS}} = 1.0 \text{ pm} = 1.0 \times 10^{-12} \text{ m}. \tag{42}$$

The controlling fabrication stack is specified in Table VI.

The Layer-20 airbridge is represented as a doubly clamped Euler–Bernoulli beam. Its unstressed and compression-shifted modal frequencies are

$$\begin{aligned}
 f_0 &= \frac{\beta_1^2 t}{2\pi\sqrt{12} L^2} \sqrt{\frac{E}{\rho}}, \\
 \beta_1 &= 4.73004, \quad t = 1.50 \text{ } \mu\text{m}, \quad L = 5.00 \text{ } \mu\text{m}, \\
 f_{\text{res}} &= f_0 \sqrt{1 - \frac{\sigma_0}{\sigma_{\text{crit}}}}, \\
 \sigma_{\text{crit}} &= \frac{\pi^2 E t^2}{3L^2}.
 \end{aligned} \tag{43}$$

Using the niobium-dominated unstressed mode $f_0 = 215.88 \text{ MHz}$, the required compressive prestress ratio is

$$\begin{aligned}
 \frac{\sigma_0}{\sigma_{\text{crit}}} &= 1 - \left(\frac{f_{\text{res}}}{f_0}\right)^2 \\
 &= 1 - \left(\frac{19.82}{215.88}\right)^2 = 0.991571.
 \end{aligned} \tag{44}$$

B. Closed-Loop PID Bias Regulation and Phase-Jitter Enforcement

Each HBT phase channel is regulated by a continuous-time proportional–integral–derivative compensator

$$\begin{aligned}
 C(s) &= K_p + \frac{K_i}{s} + K_d s \\
 &= \frac{K_d s^2 + K_p s + K_i}{s}, \\
 K_p &= 1.85 \text{ V rad}^{-1}, \\
 K_i &= 9.12 \times 10^3 \text{ V rad}^{-1} \text{ s}^{-1}, \\
 K_d &= 3.45 \times 10^{-7} \text{ V s rad}^{-1}.
 \end{aligned} \tag{45}$$

The compensator zeros are the roots of the numerator polynomial:

$$\begin{aligned}
 s_{1,2} &= \frac{-K_p \pm \sqrt{K_p^2 - 4K_d K_i}}{2K_d}, \\
 s_1 &= -4.9342701 \times 10^3 \text{ rad s}^{-1}, \\
 s_2 &= -5.3573846 \times 10^6 \text{ rad s}^{-1}, \\
 f_1 &= \frac{|s_1|}{2\pi} = 785.3135 \text{ Hz}, \\
 f_2 &= \frac{|s_2|}{2\pi} = 852.6542 \text{ kHz}.
 \end{aligned} \tag{46}$$

For plant transfer function $P(s)$, the phase-error sensitivity and complementary sensitivity are

$$\begin{aligned}
 S(s) &= \frac{1}{1 + C(s)P(s)}, \\
 T(s) &= \frac{C(s)P(s)}{1 + C(s)P(s)}, \\
 S(s) + T(s) &= 1.
 \end{aligned} \tag{47}$$

Let $S_{\phi, \text{src}}(f)$ denote source phase-noise density and $S_{\phi, \text{dac}}(f)$ denote actuation and quantization noise. Hardware-in-the-loop acceptance requires

$$\begin{aligned}
 \sigma_\phi^2 &= \int_0^{f_{\text{Nyq}}} S_{\phi, \text{src}}(f) |S(j2\pi f)|^2 df \\
 &\quad + \int_0^{f_{\text{Nyq}}} S_{\phi, \text{dac}}(f) |T(j2\pi f)|^2 df, \\
 f_{\text{Nyq}} &= 36.0 \text{ GHz}, \\
 \sigma_\phi^2 &\leq (5.05 \times 10^{-5} \text{ rad})^2 \\
 &= 2.55025 \times 10^{-9} \text{ rad}^2.
 \end{aligned} \tag{48}$$

The deterministic peak monitor supplements the integrated-noise criterion. For all channels and every 13.8889-ps control frame, compliance is established by

$$\begin{aligned}
 \|\delta\phi[k]\|_\infty &= \max_{0 \leq q \leq 63} |\delta\phi_q[k]|, \\
 \|\delta\phi[k]\|_\infty &\leq 5.05 \times 10^{-5} \text{ rad}, \\
 \sigma_\phi &\leq 5.05 \times 10^{-5} \text{ rad}.
 \end{aligned} \tag{49}$$

Equations (48) and (49) jointly enforce both the RMS and samplewise HIL phase-jitter limits.

C. Sub-Nanosecond Acoustic Tamping Stack

The acoustic tamping path transfers the 142.08-MW field-quench transient from the sapphire waveguide through a quarter-wave nanoporous silica layer into the liquid-helium-4 bath:

$$\begin{aligned}
 P_{\text{quench, pk}} &= 142.08 \text{ MW}, \\
 Z_1 &= 44.178 \text{ MRayl}, \\
 Z_2 &= 0.0275 \text{ MRayl}, \\
 T_{\text{bath}} &= 4.20 \text{ K}.
 \end{aligned} \tag{50}$$

TABLE VI
METRIC-TRANSDUCER GDSII LITHOGRAPHIC MASK STACK

Layer	GDSII purpose	Material system	Thickness / geometry	Electrical and structural requirement
10	SUBSTRATE_INP	Semi-insulating InP (100)	350.0 μm wafer	RF-low-loss mechanical datum, cryogenic heat-spreading substrate, and HBT epitaxial support; all geometry resolved on the 1.0- μm database grid.
20	AIRBRIDGE_SPAN	Micromachined Au/InP bridge	1.5 $\mu\text{m} \times 5.0 \mu\text{m}$ span	Low-capacitance crossover with stress-shifted first flexural resonance $f_{\text{bridge}} = 19.82 \text{ MHz}$.
25	MET_NB_TRACE	Superconducting Nb	300.0 nm film	Zero-resistance bias routing below $T_c = 9.20 \text{ K}$ and direct coupling to the emergency current-shunt network.

At an unmediated sapphire–helium interface, the normal-incidence intensity reflection and transmission coefficients are

$$\begin{aligned}
 R_{\text{unmatched}} &= \left(\frac{Z_1 - Z_2}{Z_1 + Z_2} \right)^2 \\
 &= \left(\frac{44.178 - 0.0275}{44.178 + 0.0275} \right)^2 \\
 &= 0.997513 = 99.7513\%, \\
 \mathcal{T}_{\text{unmatched}} &= 1 - R_{\text{unmatched}} = 0.2487\%.
 \end{aligned} \tag{51}$$

The ideal geometric-mean impedance is 1.102223 MRayl; the flight layer is deliberately offset to 1.1512 MRayl to include cryogenic densification and bonded-interface corrections:

$$\begin{aligned}
 Z_{\text{geom}} &= \sqrt{Z_1 Z_2} = 1.102223 \text{ MRayl}, \\
 Z_m &= 1.1512 \text{ MRayl}, \\
 d_m &= 6.395 \text{ nm}, \\
 d_m &= \frac{\lambda_m}{4} = \frac{v_m}{4f_{\text{ac}}}.
 \end{aligned} \tag{52}$$

At the quarter-wave design frequency, the transformed load and matched reflection coefficient are

$$\begin{aligned}
 Z_{\text{in}} &= \frac{Z_m^2}{Z_2} = 48.1913 \text{ MRayl}, \\
 R_{\text{matched}} &= \left(\frac{Z_1 - Z_{\text{in}}}{Z_1 + Z_{\text{in}}} \right)^2 \\
 &= 1.888 \times 10^{-3} = 0.1888\%, \\
 \mathcal{T}_{\text{matched}} &= 1 - R_{\text{matched}} = 99.8112\%.
 \end{aligned} \tag{53}$$

The corresponding peak reflected and transmitted powers are

$$\begin{aligned}
 P_{\text{refl,pk}} &= R_{\text{matched}} P_{\text{quench,pk}} = 268.26 \text{ kW}, \\
 P_{\text{trans,pk}} &= \mathcal{T}_{\text{matched}} P_{\text{quench,pk}} = 141.812 \text{ MW}.
 \end{aligned} \tag{54}$$

The transient pressure acceptance values and material margins are summarized in Table VII.

The pressure attenuation and governing InP safety factor are

$$\begin{aligned}
 \mathcal{A}_P &= 1 - \frac{P_{\text{InP}}}{P_{\text{sapphire}}} = 1 - \frac{8.68}{12.64} = 31.329\%, \\
 \Delta P_{\text{margin}} &= Y_{\text{InP}} - P_{\text{InP}} \\
 &= 10.00 \text{ GPa} - 8.68 \text{ GPa} = 1.32 \text{ GPa}, \\
 \mathcal{M}_Y &= \frac{\Delta P_{\text{margin}}}{Y_{\text{InP}}} = 13.20\%.
 \end{aligned} \tag{55}$$

D. Thermomechanical Stress and Coffin–Manson Fatigue

A three-dimensional Chaboche model resolves cyclic thermoviscoplastic strain across the alumina/InP/Nb interfaces. Total strain is decomposed according to

$$\begin{aligned}
 \boldsymbol{\varepsilon} &= \boldsymbol{\varepsilon}^e + \boldsymbol{\varepsilon}^p + \boldsymbol{\varepsilon}^{\text{th}}, \\
 \boldsymbol{\sigma} &= \mathbb{C} : (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^p - \boldsymbol{\varepsilon}^{\text{th}}), \\
 \boldsymbol{\varepsilon}^{\text{th}}(T) &= \int_{T_0}^T \alpha_T(\vartheta) d\vartheta \mathbf{1}, \\
 \Delta T &= 15.0 \text{ K}.
 \end{aligned} \tag{56}$$

The thermal-expansion mismatch inputs are

$$\begin{aligned}
 \alpha_{\text{Al}_2\text{O}_3} &= 5.4 \times 10^{-6} \text{ K}^{-1}, \\
 \alpha_{\text{InP}} &= 4.6 \times 10^{-6} \text{ K}^{-1}, \\
 \alpha_{\text{Nb}} &= 7.3 \times 10^{-6} \text{ K}^{-1}.
 \end{aligned} \tag{57}$$

With deviatoric stress $\mathbf{s} = \text{dev}(\boldsymbol{\sigma})$, two kinematic backstress tensors $\mathbf{X}_j$, and isotropic hardening R , the yield surface and viscoplastic flow rule are

$$\begin{aligned}
 \boldsymbol{\xi} &= \mathbf{s} - \sum_{j=1}^2 \mathbf{X}_j, \\
 f &= \sqrt{\frac{3}{2} \boldsymbol{\xi} : \boldsymbol{\xi}} - (\sigma_y + R), \\
 \dot{p} &= \left\langle \frac{f}{K} \right\rangle^n, \\
 \dot{\boldsymbol{\varepsilon}}^p &= \dot{p} \frac{3}{2} \frac{\boldsymbol{\xi}}{\sqrt{\frac{3}{2} \boldsymbol{\xi} : \boldsymbol{\xi}}}.
 \end{aligned} \tag{58}$$

The Chaboche internal variables evolve as

$$\begin{aligned}
 \dot{\mathbf{X}}_j &= \frac{2}{3} C_j \dot{\boldsymbol{\varepsilon}}^p - \gamma_j \mathbf{X}_j \dot{p} + \frac{\partial \mathbf{X}_j}{\partial T} \dot{T}, \quad j \in \{1, 2\}, \\
 \dot{R} &= b(Q - R) \dot{p}, \\
 \dot{p} &= \sqrt{\frac{2}{3} \dot{\boldsymbol{\varepsilon}}^p : \dot{\boldsymbol{\varepsilon}}^p}.
 \end{aligned} \tag{59}$$

Low-cycle fatigue is closed with the Coffin–Manson relation. For a calibrated fatigue ductility exponent $c = -0.5000$, the specified strain range and cycle life imply

$$\begin{aligned}
 \frac{\Delta \varepsilon_p}{2} &= \varepsilon'_f (2N_f)^c, \\
 \Delta \varepsilon_p &= 6.0 \times 10^{-6}, \\
 N_f &= 4.0 \times 10^6 \text{ cycles}, \\
 \varepsilon'_f &= \frac{\Delta \varepsilon_p}{2} (2N_f)^{-c} = 8.4853 \times 10^{-3}.
 \end{aligned} \tag{60}$$

TABLE VII
SUB-NANOSECOND ACOUSTIC PRESSURE TRANSMISSION AND YIELD MARGINS

Interface	E (GPa)	P_{pk} (GPa)	Y (GPa)	Margin (GPa)	Disposition
Single-crystal sapphire	400.00	12.64	22.00	9.36	Input waveguide remains below the compressive acceptance limit.
Nanoporous silica aerogel	0.05	0.31	0.85	0.54	Quarter-wave layer remains within the bonded porous-medium allowables.
Semi-insulating InP interface	61.00	8.68	10.00	1.32	Governing interface; positive 13.20% yield margin.

The mechanical cycle life maps to the de-rendering allocation as

$$\begin{aligned}
 B_{\text{life}} &= 1.514 \times 10^{16} \text{ bit}, \\
 b_{\text{cycle}} &= \frac{B_{\text{life}}}{N_f} \\
 &= \frac{1.514 \times 10^{16}}{4.0 \times 10^6} = 3.7850 \times 10^9 \text{ bit cycle}^{-1}.
 \end{aligned} \tag{61}$$

E. Fatigued Impedance and Post-Shunt Quench Margin

At the fatigue limit, aerogel densification increases the matching-layer impedance to 1.3250 MRayl. The resulting quarter-wave input impedance and reflected-power fraction are

$$\begin{aligned}
 Z_{\text{fatigue}} &= 1.3250 \text{ MRayl}, \\
 Z_{\text{in,fatigue}} &= \frac{Z_{\text{fatigue}}^2}{Z_2} = 63.8909 \text{ MRayl}, \\
 R_{\text{fatigue}} &= \left(\frac{Z_1 - Z_{\text{in,fatigue}}}{Z_1 + Z_{\text{in,fatigue}}} \right)^2 \\
 &= 0.033136 = 3.3136\%.
 \end{aligned} \tag{62}$$

Relative to the initially transmitted quench power, the reflected-power escalation is

$$\begin{aligned}
 \Delta P_{\text{refl,rel}} &= \frac{R_{\text{fatigue}} - R_{\text{matched}}}{1 - R_{\text{matched}}} \\
 &= \frac{0.033136 - 0.001888}{1 - 0.001888} \\
 &= 0.031307 = 3.13\%.
 \end{aligned} \tag{63}$$

The InP volumetric heat capacity follows the low-temperature Debye law

$$\begin{aligned}
 C_{v,\text{vol}}(T) &= a_{\text{InP}} T^3, \\
 a_{\text{InP}} &= 3.8776 \text{ J m}^{-3} \text{ K}^{-4}.
 \end{aligned} \tag{64}$$

The energy density required to heat the substrate from the helium-bath temperature to the niobium quench acceptance temperature is

$$\begin{aligned}
 Q_{\text{quench}} &= \int_{T_{\text{bath}}}^{T_{\text{quench}}} a_{\text{InP}} T^3 dT \\
 &= \frac{a_{\text{InP}}}{4} (T_{\text{quench}}^4 - T_{\text{bath}}^4), \\
 T_{\text{bath}} &= 4.2000 \text{ K}, \\
 T_{\text{quench}} &= 9.3000 \text{ K}, \\
 Q_{\text{quench}} &= 6949.97 \text{ J m}^{-3}.
 \end{aligned} \tag{65}$$

TABLE VIII
CRYOGENIC QUENCH-STATE MARGINS

State	T (K)	Headroom (K)
He-4 base bath	4.2000	5.1000
Post-shunt peak	6.8638	2.4362
Nb quench acceptance	9.3000	0.0000

For a 2.50-ns current-shunt deposition of $Q_{\text{shunt}} = 1850.00 \text{ J m}^{-3}$, inversion of the Debye integral gives

$$\begin{aligned}
 T_{\text{peak}} &= \left(T_{\text{bath}}^4 + \frac{4Q_{\text{shunt}}}{a_{\text{InP}}} \right)^{1/4} \\
 &= \left[(4.2000)^4 + \frac{4(1850.00)}{3.8776} \right]^{1/4} \text{ K} \\
 &= 6.8638 \text{ K}.
 \end{aligned} \tag{66}$$

The positive niobium quench headroom is therefore

$$\begin{aligned}
 \Delta T_{\text{headroom}} &= T_{\text{quench}} - T_{\text{peak}} \\
 &= 9.3000 \text{ K} - 6.8638 \text{ K} \\
 &= 2.4362 \text{ K} > 0.
 \end{aligned} \tag{67}$$

Table VIII closes the transient thermal acceptance chain.

IV. MODULAR LANR POWER INFRASTRUCTURE & FAULT-TOLERANT $N - k$ DERATING PROTOCOLS

A. $N + 14$ Redundant LANR Power Topology & Bus Management

The installed LANR power plant comprises $N_{\text{total}} = 1800$ independently isolatable modules connected through segmented, cross-strapped dc buses. With the verified module output $P_m = 507.3195 \text{ W}$ and thermoelectric conversion efficiency $\eta_{\text{TEG}} = 33.804\%$, the nominal net plant output and reserve above the continuous entropy-debt load are

$$\begin{aligned}
 P_{\text{net},N+14} &= N_{\text{total}} P_m \\
 &= (1800)(507.3195 \text{ W}) \\
 &= 913.1751 \text{ kW} \\
 &\simeq 913.176 \text{ kW}, \\
 \Delta P_{N+14} &= P_{\text{net},N+14} - P_{\text{debt}} \\
 &= 913.176 \text{ kW} - 906.000 \text{ kW} \\
 &= +7.176 \text{ kW}, \\
 \mathcal{M}_P &= 100 \frac{\Delta P_{N+14}}{P_{\text{debt}}} \\
 &= +0.792\%.
 \end{aligned} \tag{68}$$

The absolute module floor that sustains the full-power seed is

$$N_{\min} = \left\lceil \frac{906000.0000 \text{ W}}{507.3195 \text{ W module}^{-1}} \right\rceil \quad (69)$$

$$= 1786, \quad N_{\text{total}} = N_{\min} + 14.$$

The bus-management controller isolates a failed module, closes the appropriate cross-strap, and rebalances channel current without interrupting the non-sheddable metric load. Table IX defines the operational states from the installed $N + 14$ topology through the $N + 0$ absolute minimum. Every state at or above $N + 0$ maintains $M_{\text{seed}} = 10^{-6} M_{\odot}$ and $f_0 = 169.30 \text{ m}$; outages from $N + 13$ through $N + 1$ therefore require neither seed-mass derating nor Sensor Craft repositioning.

B. Entropy-Debt Power Requirement

The power infrastructure shall continuously compensate the entropy debt associated with the commanded ghost-seed mass. The governing proportional law is

$$P_{\text{debt}}(M_{\text{seed}}) = \left(\frac{M_{\text{seed}}}{M_{\odot}} \right) (906 \times 10^9 \text{ W}), \quad (70)$$

$$P_{\text{debt}}(10^{-6} M_{\odot}) = 906.00 \text{ kW}.$$

This load is treated as continuous and non-sheddable while the metric perturbation is active. The generation bus shall therefore provide at least 906 000 W net after all module-local pumps, chillers, controls, and conversion losses have been charged against gross production.

C. Modular LANR Cold-Fusion Cluster

Each verified SHBT LANR module combines a 2911.40-W fusion source with 182.04 W of absorbed optical power. Its gross thermal balance is

$$P_{\text{thermal}} = P_{\text{fusion}} + P_{\text{opt,abs}} \quad (71)$$

$$= 2911.40 \text{ W} + 182.04 \text{ W}$$

$$= 3093.44 \text{ W}.$$

A dual-stage skutterudite/half-Heusler thermoelectric generator (TEG) converts this heat at a verified efficiency of $\eta_{\text{TEG}} = 0.33804$:

$$P_{\text{TEG,elec}} = \eta_{\text{TEG}} P_{\text{thermal}} \quad (72)$$

$$= (0.33804)(3093.44 \text{ W})$$

$$= 1045.7065 \text{ W}.$$

The complete parasitic ledger is $P_{\text{house}} = 538.3870 \text{ W}$ per module, including the 27.24-W primary circulation pump and the 58.30-W auxiliary chiller. The remaining 452.8470 W covers reactor control, secondary circulation, power conditioning, instrumentation, and module thermal management. Consequently,

$$P_{\text{net,complete}} = P_{\text{TEG,elec}} - P_{\text{house}} \quad (73)$$

$$= 1045.7065 \text{ W} - 538.3870 \text{ W}$$

$$= +507.3195 \text{ W module}^{-1}.$$

The minimum full-seed count and installed-cluster output are therefore

$$N_{\min} = \left\lceil \frac{P_{\text{debt}}}{P_{\text{net,complete}}} \right\rceil \quad (74)$$

$$= \left\lceil \frac{906000 \text{ W}}{507.3195 \text{ W module}^{-1}} \right\rceil = 1786,$$

$$P_{\text{cluster,net}} = (1800)(507.3195 \text{ W})$$

$$= 913.1751 \text{ kW} \simeq 913.176 \text{ kW}.$$

The installed topology provides 14 module-out capability above the absolute minimum. Seed derating and focal repositioning begin only after the active population falls below $N_{\min} = 1786$, as specified in Section IV-E.

D. Deep-Space Radiator and Thermal Balance

For the complete cluster, the gross heat production and net exported electrical power give

$$P_{\text{cluster,thermal}} = (1800)(3093.44 \text{ W}) = 5.568192 \text{ MW}, \quad (75)$$

$$P_{\text{waste}} = P_{\text{cluster,thermal}} - P_{\text{cluster,net}}$$

$$= 4.6550169 \text{ MW} \simeq 4.6550 \text{ MW}.$$

The required effective radiating area follows the Stefan-Boltzmann balance

$$A_{\text{radiator}}(T_{\text{rad}}) = \frac{P_{\text{waste}}}{\epsilon \sigma (T_{\text{rad}}^4 - T_{\text{space}}^4)}, \quad (76)$$

where $\epsilon = 0.92000$, $T_{\text{space}} = 2.7 \text{ K}$, and $\sigma = 5.670374419 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$. Table X shows the strong fourth-power sensitivity of deployed area to radiator temperature.

At the 600-K baseline, Eq. (76) requires $A_{\text{radiator}} = 688.52 \text{ m}^2$ of effective emitting surface for 4.6550 MW of continuous waste-heat rejection. Packaging shall distinguish effective emitting area from projected planform area when both faces of a radiator panel have unobstructed views to deep space.

E. Fault-Tolerant $N - k$ Power Derating and Focal Repositioning

Seed derating begins only after the fourteen installed reserve modules have been exhausted and the active population falls below $N_{\min} = 1786$. The derating index k is therefore referenced to the full-seed module floor, not to the installed $N + 14$ population:

$$N_{\text{active}}(k) = N_{\min} - k = 1786 - k, \quad (77)$$

$$P_{\text{raw}}(k) = N_{\text{active}}(k) P_m,$$

$$P_{\text{net}}(k) = P_{\text{raw}}(k) - P_{\text{aux}}(k),$$

$$\Delta P_{\text{net}}(k) = P_{\text{net}}(0) - P_{\text{net}}(k),$$

$$0 \leq k \leq 1607.$$

Here $P_{\text{aux}}(k)$ is the state-dependent bus-conditioning charge; $P_{\text{aux}}(0) = 0$ and the qualified $k = 1607$ endpoint includes a 2.0-W conditioning charge. The positive net-power loss is

TABLE IX
OPERATIONAL LANR REDUNDANCY STATES

State	Active modules	Redundant modules	Net power (kW)	Reserve (kW)	Bus-management disposition
$N + 14$ nominal	1800	14	913.176	+7.176	Baseline load distribution across all available channels.
$N + 13$ single outage	1799	13	912.6678	+6.6678	Failed module isolated; optical-drive allocation unchanged.
$N + 12$ double outage	1798	12	912.1605	+6.1605	Remaining channels absorb the load through current rebalancing.
$N + 10$ degraded	1796	10	911.1458	+5.1458	Secondary diagnostics are power-limited; metric synthesis remains fully powered.
$N + 5$ intermediate	1791	5	908.6092	+2.6092	Optical phase synthesis is prioritized over nonessential loads.
$N + 1$ critical margin	1787	1	906.5799	+0.5799	Critical-state telemetry is asserted and discretionary loads are shed.
$N + 0$ absolute minimum	1786	0	906.0726	+0.0726	Full seed mass and the 169.30-m focal baseline are maintained at the redundancy boundary.

TABLE X
RADIATOR AREA VERSUS OPERATING TEMPERATURE

T_{rad} (K)	A_{radiator} (m ²)
400	3485.63
450	2176.06
500	1427.72
550	975.15
600	688.52
650	499.88
700	371.65

At $k = 1607$, the derating base contains 179 active modules. The verified endpoint ledger is

$$\begin{aligned}
 N_{\text{active}}(1607) &= 179, \\
 P_{\text{net}}(1607) &= 90.8082 \text{ kW}, \\
 P_{\text{seed}}(1607) &= 90.6000 \text{ kW}, \\
 P_{\text{reserve}}(1607) &= 208.2 \text{ W}, \\
 M_{\text{seed}}(1607) &= 10^{-7} M_{\odot}.
 \end{aligned} \tag{81}$$

converted into the discrete decrement applied to the 512-bit command register:

$$\begin{aligned}
 \Delta N(k) &= \left\lfloor \frac{\Delta P_{\text{net}}(k)}{P_{\text{bit}}} \right\rfloor, \\
 P_{\text{bit}} &= 8.9506 \times 10^{-4} \text{ W bit}^{-1}, \\
 0 \leq \Delta N(k) &\leq 2^{512} - 1.
 \end{aligned} \tag{78}$$

For an isolated loss of one qualified module,

$$\begin{aligned}
 \Delta P_{\text{net}}(1) &= 507.3195 \text{ W}, \\
 \Delta N(1) &= \left\lfloor \frac{507.3195}{8.9506 \times 10^{-4}} \right\rfloor = 566\,799 \text{ bit}.
 \end{aligned} \tag{79}$$

The register decrement is a finite-command representation and is distinct from the microscopic boundary-state count ΔN_0 in Eq. (1).

The seed-power command is bounded by the full-power entropy debt and the qualified science-safe floor:

$$\begin{aligned}
 P_{\text{seed}}(k) &= \max[P_{\text{seed,min}}, \min(P_{\text{net}}(k), P_{\text{debt},0})], \\
 P_{\text{debt},0} &= 906.000 \text{ kW}, \\
 P_{\text{seed,min}} &= 90.60 \text{ kW}, \\
 M_{\text{seed}}(k) &= M_{\text{seed},0} \frac{P_{\text{seed}}(k)}{P_{\text{debt},0}}, \\
 M_{\text{seed,min}} &= 10^{-7} M_{\odot}.
 \end{aligned} \tag{80}$$

The fourteen installed reserve modules are outside the definition of k ; thus, the same endpoint corresponds to 1621 unavailable modules when failures are counted from the installed population of 1800.

Because the thin-lens focal distance varies inversely with seed mass, the commanded inter-craft separation is

$$\begin{aligned}
 f(k) &= f_0 \frac{M_{\text{seed},0}}{M_{\text{seed}}(k)} = f_0 \frac{P_{\text{debt},0}}{P_{\text{seed}}(k)}, \\
 f(0) &= 169.30 \text{ m}, \\
 f(1607) &= 169.30 \left(\frac{10^{-6}}{10^{-7}} \right) = 1\,692.99 \text{ m}.
 \end{aligned} \tag{82}$$

The state ledger below defines the transition from redundant full-seed operation to the qualified minimum seed.

For a commanded axial displacement $\Delta z = f(k_2) - f(k_1)$, the Sensor Craft uses normalized time $\tau = t/T_{\text{trans}}$ and the fifth-order minimum-jerk trajectory

$$\begin{aligned}
 s(\tau) &= 10\tau^3 - 15\tau^4 + 6\tau^5, \quad 0 \leq \tau \leq 1, \\
 z_{\text{ref}}(t) &= z_0 + \Delta z s(\tau).
 \end{aligned} \tag{83}$$

The dimensionless kinematic functions and exact acceptance

QUALIFIED $N - k$ SEED-DERATING AND FOCAL-REPOSITIONING STATES

k	Active modules	Seed power (kW)	$M_{\text{seed}}/M_{\odot}$	f (m)	Operational disposition
0	1786	906.000	1.0000×10^{-6}	169.30	Full seed; fourteen installed modules remain outside the derating base.
1	1785	905.565	9.9952×10^{-7}	169.38	Continuous command decrement; metrology remains closed during axial correction.
1607	179	90.600	1.0000×10^{-7}	1 692.99	Science-safe minimum; 208.2 W remains above the seed demand.

bounds are

$$\begin{aligned}
\dot{s}(\tau) &= 30\tau^2 - 60\tau^3 + 30\tau^4, \\
\ddot{s}(\tau) &= 60\tau - 180\tau^2 + 120\tau^3, \\
\max_{\tau \in [0,1]} \dot{s}(\tau) &= 1.8750 \quad \text{at } \tau = \frac{1}{2}, \\
\max_{\tau \in [0,1]} |\ddot{s}(\tau)| &= 5.7733 \quad \text{at } \tau = \frac{3 \mp \sqrt{3}}{6}, \\
|\dot{z}|_{\max} &= 1.8750 \frac{|\Delta z|}{T_{\text{trans}}}, \\
|\ddot{z}|_{\max} &= 5.7733 \frac{|\Delta z|}{T_{\text{trans}}^2}.
\end{aligned} \tag{84}$$

The endpoint conditions $s(0) = 0$, $s(1) = 1$, $\dot{s}(0) = \dot{s}(1) = 0$, and $\ddot{s}(0) = \ddot{s}(1) = 0$ eliminate commanded velocity and acceleration steps. Invalid power telemetry, disagreement between the module-validity mask and the 512-bit decrement, loss of heterodyne/DWS lock, or available seed power below 90.60 kW commands an orderly field quench rather than extrapolation below the qualified envelope.

F. Active Entropic Refrigeration

The 4.2-K boundary assembly uses active entropic refrigeration to remove the heat associated with controlled de-rendering events. If Γ_{de} is the event rate and each event removes one bit with entropy increment $\Delta S = k_B \ln 2$, the ideal cold-side cooling rate is

$$\begin{aligned}
P_{\text{cool}} &= \Gamma_{\text{de}} \Delta S T_c \\
&= \Gamma_{\text{de}} k_B T_c \ln 2, \\
T_c &= 4.2 \text{ K}, \quad k_B = 1.380649 \times 10^{-23} \text{ J K}^{-1}.
\end{aligned} \tag{85}$$

Thus, the ideal event rate required per watt of cold-side removal is

$$\frac{\Gamma_{\text{de}}}{P_{\text{cool}}} = \frac{1}{k_B T_c \ln 2} = 2.4880 \times 10^{22} \text{ bit J}^{-1}. \tag{86}$$

Let $q_b'' = P_b/A_b$ denote the net heat flux crossing boundary area A_b , and let ΔT_b be the corresponding temperature rise. The refrigeration controller shall command Γ_{de} such that

$$\begin{aligned}
P_{\text{cool}} &\geq P_b, \\
R_{\text{th},b}'' &= \frac{\Delta T_b}{q_b''} < 7.04 \times 10^{-6} \text{ m}^2 \text{ K W}^{-1}.
\end{aligned} \tag{87}$$

The control law shall use the measured boundary heat load, not the ideal entropy term alone, when setting the commanded event rate; conversion inefficiency and parasitic cryogenic loads are carried as additional cold-stage margin.

V. C-ABI ZERO-COPY SHARED MEMORY TELEMETRY ARCHITECTURE AND BARE-METAL MICROKERNEL

A. POSIX Shared-Memory SPSC Transport

High-rate telemetry is transported through two unidirectional POSIX shared memory rings: a driver-to-shbt-os acquisition ring and a shbt-os-to-user-daemon publication ring. Each object is created with `shm_open`, sized once with `ftruncate`, and mapped into the participating address spaces with `mmap(..., MAP_SHARED, ...)`. The hardware driver maps DMA destination pages directly onto producer-owned frame slots, and the consumer reads the committed slot in place. No payload copy, serialization pass, system-call transition, or heap allocation occurs in the steady-state data path.

Each ring is a power-of-two array of Q telemetry frames with monotonic producer and consumer indices

$$\begin{aligned}
h &= \text{write_head}, \quad t = \text{read_tail}, \\
n_{\text{used}} &= h - t, \quad q_{\text{quadn}}_{\text{free}} = Q - (h - t), \\
i_{\text{write}} &= h \& (Q - 1), \quad q_{\text{quad}}_{\text{read}} = t \& (Q - 1).
\end{aligned} \tag{88}$$

The producer owns `write_head`; the consumer owns `read_tail`. Both are C-ABI-compatible `AtomicUsize` values. The producer writes a complete frame and publishes $h + 1$ with release ordering. The consumer loads h with acquire ordering, validates the frame, processes it in place, and publishes $t + 1$ with release ordering. Full and empty states are defined by $h - t = Q$ and $h = t$, respectively. Unsigned monotonic-index subtraction makes wraparound well defined for ring occupancies smaller than half the index space.

The verified end-to-end transfer latency from producer release to consumer acquire is

$$\begin{aligned}
t_{\text{shm}} &= 0.561 \mu\text{s}, \\
t_{\text{shm}} &< 1.000 \mu\text{s}.
\end{aligned} \tag{89}$$

The frame type and both index records use strict cache-line layout equivalent to `#[repr(C, align(64))]`. The write-head record, read-tail record, and frame array begin on separate 64-byte cache lines. Consequently, the producer never writes the consumer's cache line and the consumer never writes the producer's cache line, eliminating false sharing and cache-line bouncing from the SPSC synchronization path.

The frame-size and alignment invariants are

$$\begin{aligned}
\text{sizeof}(\text{TelemetryFrame}) &= 64 \text{ byte}, \\
\text{alignof}(\text{TelemetryFrame}) &= 64 \text{ byte}, \\
\text{offsetof}(\text{crc16}) &= 62 \text{ byte}.
\end{aligned} \tag{90}$$

All multibyte fields use little-endian representation. A frame becomes visible only after DMA completion, sequence-number assignment, status capture, and CRC generation. A

TABLE XI
COMPLETE 64-BYTE CACHE-ALIGNED TELEMETRY FRAME LAYOUT

Byte offset	Field	C type	Representation	C-ABI function
0x00–0x07	timestamp	uint64_t	Epoch ns	Unsigned nanoseconds from the configured mission epoch.
0x08–0x27	phase_error[16]	int16_t[16]	Q15 rad	Sixteen signed optical-phase residuals; physical phase is $q/2^{15}$ rad.
0x28–0x2F	substrate_temp[4]	uint16_t[4]	Q8.8 K	Four quadrant substrate temperatures; physical temperature is $q/2^8$ K.
0x30–0x37	piezo_stress[4]	uint16_t[4]	Q8.8 MPa	Four quadrant mechanical-stress samples; physical stress is $q/2^8$ MPa.
0x38–0x3B	sequence_id	uint32_t	Unsigned counter	Monotonic frame identifier for loss, duplication, and ordering detection.
0x3C–0x3D	status_flags	uint16_t	Bit field	Encodes the $N + k$ power state, wide-angle steering state, metrology validity, ECC state, and shunt status.
0x3E–0x3F	crc16	uint16_t	CCITT-16	CRC-16/CCITT over bytes 0x00–0x3D; the receiver validates it before advancing <code>read_tail</code> .

CRC failure or discontinuous `sequence_id` marks the slot invalid without exposing partial telemetry to the deterministic control state.

B. Bare-Metal *shbt-os* Microkernel and UnifiedStine-springFrame SRAM Arena

The flight-control computer shall execute *shbt-os* as a freestanding, bare-metal C11 environment. The build shall use the freestanding implementation model, provide no heap allocator, perform no dynamic allocation after reset, and link against no hosted C library. Startup code, fixed-width arithmetic, memory primitives, interrupt entry, and MMIO access shall be supplied by the verified microkernel image. All control paths shall have statically bounded memory use and shall avoid recursion, unbounded loops, and data-dependent allocation.

The linker shall reserve one contiguous UnifiedStinespringFrame in on-chip SRAM, align its base to a 64-byte boundary, and assert its size at link time:

$$\begin{aligned}
 A_{\text{frame}} \bmod 64 &= 0, \\
 B_{\text{frame}} &= 2112 \text{ byte}, \\
 B_A &= 640 \text{ byte} = \frac{10}{33} B_{\text{frame}}, \\
 B_D &= 1472 \text{ byte} = \frac{23}{33} B_{\text{frame}}, \\
 B_A + B_D &= 2112 \text{ byte}.
 \end{aligned} \tag{91}$$

The Active Visible Register occupies the first 640 bytes and has rational dilation weight $\eta_A = 10/33$. It contains the live 64-channel state vector, threshold buffers, command shadow registers, and current-cycle ECC work products. The Dark Ledger occupies the following 1472 bytes with $\eta_D = 23/33$. Of this region, 124 fixed-width braid descriptors consume

$$B_{\text{braid}} = (124)(8 \text{ byte}) = 992 \text{ byte}, \tag{92}$$

leaving 480 bytes for ledger indices, ECC syndromes, recovery checkpoints, and deterministic padding. The identities $\eta_A + \eta_D = 1$ and $B_A : B_D = 10 : 23$ shall be compile-time invariants.

C. Normative SHBT-MMIO-1 Core Register Map

Table XII is the normative SHBT-MMIO-1 core register block. Fourteen contiguous 32-bit registers span byte offsets 0x00 through 0x34; hence the occupied interval is

$$B_{\text{MMIO}} = 0x38 = 56 \text{ byte}, \quad A_{\text{base}} = 0x70000000. \tag{93}$$

All registers are naturally aligned, strongly ordered, and excluded from caches. The interference-tensor aperture beginning at 0x70000040 is an extension window and is not included in the 56-byte core span.

D. SECDED ECC and Constant-Time Channel Remapping

Every persistent 64-bit SRAM payload shall be stored as a Hamming (72, 64) SECDED codeword. Seven Hamming check bits occupy the power-of-two positions in the first 71 codeword locations, and the seventy-second bit supplies overall parity. For received codeword c , the decoder forms the seven-bit syndrome s and overall-parity mismatch p and applies

$$\begin{aligned}
 s = 0, p = 0 &\Rightarrow \text{no error}, \\
 s \neq 0, p = 1 &\Rightarrow \text{correct one indexed bit}, \\
 s = 0, p = 1 &\Rightarrow \text{correct the overall-parity bit}, \\
 s \neq 0, p = 0 &\Rightarrow \text{detect double-bit error}.
 \end{aligned} \tag{94}$$

Correction shall occur before a payload enters the Active Visible Register. A double-bit detection shall inhibit the affected command, latch `ECC_SYNDROME`, and invoke channel replacement rather than release corrupted state to the transducer array.

Dynamic replacement is performed by `shbt_remap`, which applies an AVX-512 Givens rotation to the failed and spare channel components. For channel indices i and j ,

$$\begin{bmatrix} x'_i \\ x'_j \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x_i \\ x_j \end{bmatrix}. \tag{95}$$

The selected coefficient pair is loaded from a fixed table, and masked permutes replace the selected lanes without scanning the channel set. One vector load, one fixed shuffle/multiply-add schedule, and one vector store therefore give $O(1)$ execution with respect to the number of active channels. The

TABLE XII
NORMATIVE SHBT-MMIO-1 CORE REGISTER MAP

Address	Symbol	Access	Width	Normative function
0x70000000	CR0_CTRL	R/W	32 bit	Core enable, operating-mode selection, phase-loop arm, and synchronous soft reset.
0x70000004	SR0_STAT	R	32 bit	Phase-lock, thermal-alarm, metrology-valid, ECC, quench, and shunt-latched status.
0x70000008	DET_MU_HI	R	32 bit	Upper word of signed Q1.62 eigenvector-detuning telemetry.
0x7000000C	DET_MU_LO	R	32 bit	Lower word of signed Q1.62 eigenvector-detuning telemetry.
0x70000010	BIT_OVF_N	R/W	32 bit	Upper command word of the 64-bit boundary-state overflow mantissa.
0x70000014	BIT_OVF_L	R/W	32 bit	Lower command word of the 64-bit boundary-state overflow mantissa.
0x70000018	RF_PHASE_V	R/W	32 bit	Channel address and phase-bias DAC command.
0x7000001C	SHUNT_TRIG	W	32 bit	Any nonzero write asserts the emitter-current diversion network; a zero write has no effect.
0x70000020	REC_SEQ_PC	R/W	32 bit	Write loads the recovery entry point and starts execution; read returns the microsequence PC and terminal code.
0x70000024	ECC_SYNDROME	R	32 bit	Latest SECEDED syndrome, corrected-bit index, and single-/double-error classification.
0x70000028	ECC_CORR_ADDR	R	32 bit	SRAM word address associated with the most recent corrected or uncorrectable codeword.
0x7000002C	REMAP_SRC	R/W	32 bit	Failed logical-channel identifier and remapping generation number.
0x70000030	REMAP_DST	R/W	32 bit	Replacement physical-channel identifier and Givens-rotation slot.
0x70000034	IRQ_MASK_ACK	R/W	32 bit	Interrupt mask on write and latched-source acknowledgement on write-one-to-clear.

routine shall preserve the vector norm to the fixed-point rounding bound and shall commit `REMAP_SRC` and `REMAP_DST` before the replacement channel is enabled.

E. AVX-512 SIMD Safety Interlock and Post-Quench Recovery

The highest-criticality monitor operates on two 64-byte stack buffers whose addresses satisfy $A \bmod 64 = 0$. One holds sixteen lane-wise detuning magnitudes and the other holds the replicated 10^{-12} threshold. The normative comparison pipeline is

$$\text{vmovaps} \rightarrow \text{vcmppps} \rightarrow \text{vmovmskps} \rightarrow \text{mov}, \quad (96)$$

where the compare is true for any lane satisfying $|\delta\mu_i| > 10^{-12}$. The resulting nonzero mask is written directly to `SHUNT_TRIG`; because a zero write is defined as a no-operation, the MMIO assertion path contains no conditional branch. Four vector groups cover all 64 channels through a fixed unrolled schedule and feed a hardware OR reduction before the single MMIO write.

Cycle-accurate hardware-in-the-loop timing of the aligned load, compare, mask extraction, reduction, and strongly ordered MMIO write gives

$$\begin{aligned} t_{\text{SIMD}} &= 1.412 \text{ ns}, \\ t_{\text{shunt,max}} &= 2.500 \text{ ns}, \\ t_{\text{arithmetic margin}} &= t_{\text{shunt,max}} - t_{\text{SIMD}} = 1.088 \text{ ns}, \\ t_{\text{safety margin}} &= 1.00 \text{ ns}, \\ t_{\text{bridge allowance}} &= 0.088 \text{ ns}. \end{aligned} \quad (97)$$

The MMIO bridge shall provide a deterministic bypass path such that a nonzero mask reaches `0x7000001C` and asserts the current shunt before 2.50 ns from the sampled threshold violation. The certified 1.00-ns margin is retained after reserv-

ing 0.088 ns for bridge skew and aperture uncertainty. The governing predicate is

$$\max_i |\delta\mu_i| > 10^{-12} \implies \text{SHUNT_TRIG} \leftarrow 0xFFFFFFFF. \quad (98)$$

Post-quench recovery.

Following shunt assertion, hardware holds all phase-bias outputs below turn-on for 110.00 ns while the acoustic and niobium transients decay. Once thermal-valid, phase-valid, and ECC-valid are simultaneously asserted, `shbt-os` writes the fixed `shbt_recover` entry point to `REC_SEQ_PC` at `0x70000020`. The sequencer restores the last ECC-clean checkpoint, applies any pending channel remap, and re-enables the phase loop in 9.24 ns:

$$\begin{aligned} t_{\text{hold}} &= 110.00 \text{ ns}, \\ t_{\text{sequence}} &= 9.24 \text{ ns}, \\ t_{\text{post-quench}} &= t_{\text{hold}} + t_{\text{sequence}} \\ &= 119.24 \text{ ns} < 120.00 \text{ ns}, \\ t_{\text{recovery margin}} &= 0.76 \text{ ns}. \end{aligned} \quad (99)$$

Any failed validity check, uncorrectable SECEDED event, or renewed rigidity violation leaves the shunt latched and prevents the recovery PC from advancing to the phase-enable terminal state.

VI. IN-FLIGHT GNC, DUAL-WAVELENGTH LASER METROLOGY, AND NON-PARAXIAL OPTICS

A. Dual-Wavelength Heterodyne Laser Interferometry and DWS

The inter-craft metrology system combines two frequency-stabilized optical carriers,

$$\lambda_1 = 1064.500 \text{ nm}, \quad \lambda_2 = 1064.520 \text{ nm}, \quad (100)$$

to provide optical-fringe precision together with a synthetic wavelength for integer-fringe resolution. The resulting synthetic wavelength is

$$\lambda_{\text{syn}} = \frac{\lambda_1 \lambda_2}{|\lambda_1 - \lambda_2|} = 56.6583 \text{ mm}. \quad (101)$$

Balanced InP/InGaAs photodetectors reject common-mode intensity noise and downconvert the received optical fields to an 80.0-MHz intermediate-frequency beat. A digital phase-locked loop unwraps the synthetic-wavelength phase and then estimates fine range from the carrier phase,

$$\Phi_{\text{het}}(t) = \frac{4\pi}{\lambda_1} r(t) + \phi_{\text{inst}}(t) + \phi_{\text{noise}}(t). \quad (102)$$

For $f > 1.0$ Hz, the one-sided phase-noise amplitude spectral density shall satisfy

$$S_{\phi}^{1/2}(f) \leq 2.0 \text{ } \mu\text{rad}/\sqrt{\text{Hz}}, \quad (103)$$

which establishes the allocated single-axis range-noise limit

$$\begin{aligned} S_{r,\text{lim}}^{1/2}(f) &= \frac{\lambda_1}{4\pi} S_{\phi}^{1/2}(f) \\ &\leq 0.170 \text{ pm}/\sqrt{\text{Hz}}, \\ S_{r,\text{ver}}^{1/2}(f) &= 0.144 \text{ pm}/\sqrt{\text{Hz}}. \end{aligned} \quad (104)$$

The verified closed-loop formation result is

$$\|\delta \mathbf{r}\|_{3\sigma} = 0.772 \text{ nm} < 1.000 \text{ nm}. \quad (105)$$

Bias calibration, cyclic nonlinearity, clock transfer, beam walk, and thermoelastic drift are separately estimated in the state filter over the 169.30-m reference baseline.

Angular knowledge is obtained by differential wavefront sensing (DWS) on redundant quadrant-photodiode (QPD) arrays. Opposed-quadrant heterodyne phases form pitch and yaw observables,

$$e_{\theta,x} = K_{\text{DWS}}^{-1}(\bar{\Phi}_A - \bar{\Phi}_B), \quad e_{\theta,y} = K_{\text{DWS}}^{-1}(\bar{\Phi}_C - \bar{\Phi}_D), \quad (106)$$

where K_{DWS} is calibrated against beam size, curvature, and QPD geometry. The verified fused attitude solution is

$$\sigma_{\theta,\text{ver}} = 11.38 \text{ nrad} < 15.00 \text{ nrad} = \sigma_{\theta,\text{lim}}, \quad (107)$$

with sub-nanoradian readout resolution retained for control-law quantization and calibration.

B. Sun–Earth L2 Environmental Disturbance Modeling

Solar-radiation pressure (SRP) on either craft is modeled by

$$\mathbf{a}_{\text{SRP}} = -\frac{C_r S_0}{c} \frac{A}{m} \left(\frac{1 \text{ AU}}{R} \right)^2 \hat{\mathbf{s}}, \quad (108)$$

where $S_0 = 1361 \text{ W m}^{-2}$, C_r is the effective radiation-pressure coefficient, and $\hat{\mathbf{s}}$ points toward the Sun. The Lens Craft and Sensor/Focal Craft area-to-mass ratios, approximately 0.0150 and 0.0042 $\text{m}^2 \text{ kg}^{-1}$, respectively, produce the modeled differential acceleration

$$\Delta \mathbf{a}_{\text{SRP}} \approx 3.42 \times 10^{-8} \text{ m s}^{-2} \hat{\mathbf{s}}. \quad (109)$$

The linearized Sun–Earth gravity-gradient contribution is propagated with the time-varying tidal tensor evaluated along the halo-orbit reference trajectory,

$$\begin{aligned} \mathbf{a}_{\text{grav,diff}}(t) &= \mathbf{C}_{\text{tidal}}(t) \delta \mathbf{r}(t), \\ \mathbf{C}_{\text{tidal}} &= \sum_{j \in \{\odot, \oplus\}} \frac{GM_j}{R_j^3} \left(3\hat{\mathbf{R}}_j \hat{\mathbf{R}}_j^T - \mathbf{I} \right). \end{aligned} \quad (110)$$

At the nominal L2 geometry, the combined Sun–Earth tensor produces an axial differential acceleration of order $4.7 \times 10^{-11} \text{ m s}^{-2}$ over 169.30 m; the flight propagator shall use ephemeris-resolved $\mathbf{C}_{\text{tidal}}(t)$ rather than this scalar design value.

Residual outgassing is represented by a decaying bias plus stochastic acceleration,

$$a_{\text{out}}(t) = a_{0,\text{out}} e^{-t/\tau_{\text{out}}} + \sqrt{2D_{\text{out}}} \xi(t), \quad \tau_{\text{out}} = 14.0 \text{ day}, \quad (111)$$

with $S_{a,\text{out}}^{1/2}(f) \leq 1.0 \times 10^{-10} \text{ m s}^{-2}/\sqrt{\text{Hz}}$. Thermoelastic metrology bias follows

$$\Delta L(t) = \alpha_{\text{CTE}} L_b \Delta T(t), \quad \alpha_{\text{CTE}} = 1.2 \times 10^{-7} \text{ K}^{-1}, \quad (112)$$

where the $L_b = 3.75$ -mm effective C/SiC bench path yields 0.45 nm K^{-1} . Common-mode expansion of the inter-craft baseline is not substituted for this internal optical-path coefficient.

C. Ten-Year GNC Propellant Consumption Budget

The primary mission interval is $T_{10} = 10.000 \text{ yr}$. Both craft use EMI-BF₄ ionic-liquid electrospray propulsion with $I_{\text{sp}} = 3200.0 \text{ s}$. The installed propellant capacities are 850.0 kg for the Lens Craft and 340.0 kg for the Sensor Craft. The propulsion ledger uses beginning-of-life maneuvering masses $m_{\text{L},0} = 4250.0 \text{ kg}$ and $m_{\text{S},0} = 1700.0 \text{ kg}$ for the two respective control plants.

The annual differential-SRP increment follows directly from Eq. (109):

$$T_{\text{yr}} = 365.25(86400 \text{ s}) = 31\,557\,600 \text{ s},$$

$$\begin{aligned} \Delta v_{\text{SRP,yr}} &= \Delta a_{\text{SRP}} T_{\text{yr}} \\ &= (3.42 \times 10^{-8} \text{ m s}^{-2})(31\,557\,600 \text{ s}) \\ &= 1.0792 \text{ m s}^{-1} \text{ yr}^{-1}, \end{aligned} \quad (113)$$

$$\Delta v_{\text{SRP,10}} = 10.792 \text{ m s}^{-1}.$$

The outgassing compensation allocation is based on the conservative secular acceleration $a_{\text{outgas}} = 5.0 \times 10^{-9} \text{ m s}^{-2}$:

$$\begin{aligned} \Delta v_{\text{outgas,yr}} &= a_{\text{outgas}} T_{\text{yr}} = 0.1578 \text{ m s}^{-1} \text{ yr}^{-1}, \\ \Delta v_{\text{outgas,10}} &= 1.578 \text{ m s}^{-1}. \end{aligned} \quad (114)$$

Halo-orbit station keeping contributes $\Delta v_{\text{sk}} = 2.5000 \text{ m s}^{-1} \text{ yr}^{-1}$, or 25.000 m s^{-1} over ten years. Autonomous focal transitions carry separate ten-year allowances of 15.000 m s^{-1} for the Lens Craft and 25.000 m s^{-1} for the Sensor Craft.

The corresponding propellant expenditure is evaluated with the Tsiolkovsky relation

$$\Delta m = m_0 \left[1 - \exp\left(-\frac{\Delta v}{g_0 I_{\text{sp}}}\right) \right], \quad g_0 = 9.80665 \text{ m s}^{-2}. \quad (115)$$

TABLE XIII
SUN-EARTH L2 FORMATION-DISTURBANCE ENVIRONMENT

Disturbance	Magnitude	Frequency domain / characteristic	Principal GNC effect and accommodation
Differential SRP	$3.42 \times 10^{-8} \text{ m s}^{-2}$	Secular; $f < 10^{-4} \text{ Hz}$	Axial drift and transverse bias; feed-forward SRP model plus electrospray trim.
Sun-Earth tidal gravity	$\sim 4.7 \times 10^{-11} \text{ m s}^{-2}$ at 169.30 m	Halo-orbit harmonics; fundamental near $6.6 \times 10^{-7} \text{ Hz}$	Periodic baseline dilation and attitude coupling; ephemeris-resolved state propagation.
Residual outgassing	$S_a^{1/2} \leq 1.0 \times 10^{-10} \text{ m s}^{-2}/\sqrt{\text{Hz}}$	Broadband, 10^{-5} – 10^{-1} Hz ; 14.0-day decay	Stochastic translation and torque; augmented-state bias estimation and low-frequency thrust rejection.
C/SiC thermoelastic path	0.45 nm K^{-1}	Thermal transients, 10^{-4} – 10^{-2} Hz	Metrology phase bias; temperature feed-forward and in-flight zero-path calibration.
Mechanism and electrical jitter	Allocated from measured line spectra	0.1 Hz–100 kHz	Beam-pointing and phase modulation; electro-optic correction without spacecraft momentum exchange.

TABLE XIV
TEN-YEAR CUMULATIVE GNC DELTA-V BUDGET

Contributor	Lens annual (m/s/yr)	Lens 10-yr (m/s)	Sensor annual (m/s/yr)	Sensor 10-yr (m/s)
Halo-orbit station keeping	2.5000	25.000	2.5000	25.000
Differential solar-radiation-pressure rejection	1.0792	10.792	1.0792	10.792
Spacecraft outgassing compensation	0.1578	1.578	0.1578	1.578
Autonomous focal repositioning allowance	—	15.000	—	25.000
Total cumulative allocation	3.7370 recurring	52.370	3.7370 recurring	62.370
EMI-BF ₄ propellant consumed (kg)	—	7.0900	—	3.3800
Installed propellant capacity (kg)	—	850.0	—	340.0
End-of-life reserve margin (%)	—	99.17	—	99.01

For the cumulative allocations in Table XIV,

$$\begin{aligned}
 \Delta m_{\text{lens}} &= (4250.0 \text{ kg}) \left[1 - \exp\left(-\frac{52.370}{(3200.0)(9.80665)}\right) \right] \\
 &= 7.0900 \text{ kg}, \\
 \Delta m_{\text{sensor}} &= (1700.0 \text{ kg}) \left[1 - \exp\left(-\frac{62.370}{(3200.0)(9.80665)}\right) \right] \\
 &= 3.3800 \text{ kg}.
 \end{aligned} \tag{116}$$

The end-of-life propellant inventories and reserve fractions are

$$\begin{aligned}
 m_{\text{L,EOL}} &= 850.0 - 7.0900 = 842.9100 \text{ kg}, \\
 \mathcal{R}_{\text{L,EOL}} &= 99.17\%, \\
 m_{\text{S,EOL}} &= 340.0 - 3.3800 = 336.6200 \text{ kg}, \\
 \mathcal{R}_{\text{S,EOL}} &= 99.01\%.
 \end{aligned} \tag{117}$$

Both propulsion systems therefore retain more than 98% of installed propellant at the end of the ten-year primary mission.

Focal-repositioning commands use the fifth-order minimum-jerk trajectory and the velocity/acceleration bounds specified by Eqs. (83) and (84) in Section IV-E.

D. Two-Tier Hybrid Control Architecture

The relative-state estimator fuses heterodyne range, DWS pitch and yaw, optical-clock transfer, and inertial propagation. Complementary filters divide the range and pointing residuals at the 0.1-Hz crossover. Low-frequency errors are assigned to multimodal EMI-BF₄ ionic-liquid electrospray microthrusters

with a commanded range of 0.1–150.0 μN and $I_{\text{sp}} = 3200.0 \text{ s}$. The thrust allocator removes secular SRP, gravity-gradient, and outgassing biases while maintaining the electro-optic stage near midrange.

For $0.1 \text{ Hz} \leq f \leq 100 \text{ kHz}$, an integrated 8×8 InP/InGaAs phase-shifter array corrects optical-path and beam-pointing jitter without imparting mechanical momentum. Each channel obeys

$$\begin{aligned}
 \Delta\phi_{\text{EO}}(V) &= \frac{\pi}{V_{\pi}} [V_{\text{bias}}(t) - V_0], \\
 V_{\pi} &= 3.6 \text{ V}, \quad V_0 = 3.8 \text{ V},
 \end{aligned} \tag{118}$$

with $3.8 \leq V_{\text{bias}} \leq 7.4 \text{ V}$, response time no greater than 5 ns, and device modulation bandwidth above 40 GHz. The control band is limited to 100 kHz by metrology latency and noise shaping rather than device speed.

For measured range error $e_r = r_{\text{cmd}} - r_{\text{meas}}$, the complementary command split is

$$\begin{aligned}
 u_{\text{thr}}(s) &= G_{\text{thr}}(s)H_{\text{LP}}(s)e_r(s), \\
 u_{\text{EO}}(s) &= G_{\text{EO}}(s)H_{\text{HP}}(s)e_r(s),
 \end{aligned} \tag{119}$$

where $H_{\text{LP}} + H_{\text{HP}} = 1$ and both paths include saturation and anti-windup logic. The electro-optic array provides a calibrated $\pm 10 \mu\text{m}$ equivalent optical-path stroke. Persistent stroke utilization above 80% transfers bias to the thruster loop; QPD loss of lock freezes the phase command and initiates a low-rate spiral reacquisition rather than integrating an unobservable error.

Power-fault handling, 512-bit command decrements, continuous seed-mass derating, and Sensor-Craft focal repositioning are governed by Section IV-E. The formation controller receives the authenticated module-validity mask, seed-power command, register decrement, seed-mass command, and command epoch and rejects stale or internally inconsistent tuples before issuing any axial trajectory.

E. Metrology-Guided Formation State Estimation and Optical-Axis Control

The flight estimator propagates the relative translational state, DWS attitude error, clock offset, and internal optical-path bias in a single covariance state,

$$\mathbf{x}_{\text{GNC}} = [\delta \mathbf{r}^\top \quad \delta \mathbf{v}^\top \quad \delta \boldsymbol{\theta}^\top \quad \delta t_{\text{mathrm{mclk}}} \quad \delta L_{\text{int}}]^\top. \quad (120)$$

The commanded Sensor-Craft position is constrained to the instantaneous optical axis while retaining transverse trim authority:

$$\begin{aligned} \mathbf{r}_{\text{SF,cmd}}(t) &= \mathbf{r}_{\text{LC}}(t) + f(t)\hat{\mathbf{k}}(t) \\ &\quad + \mathbf{P}_\perp(t)\delta \mathbf{r}_\perp(t), \\ \mathbf{P}_\perp &= \mathbf{I} - \hat{\mathbf{k}}\hat{\mathbf{k}}^\top. \end{aligned} \quad (121)$$

Heterodyne range closes the axial degree of freedom, DWS closes pitch and yaw, and the electro-optic stage removes residual high-frequency path error before the non-paraxial phase command is evaluated.

F. Non-Paraxial Wide-Angle Wavefront Optics & DAC Pre-Distortion

Tracking commands with $5^\circ < \Theta_{\text{tilt}} \leq 15^\circ$ use an exact eikonal path model rather than the paraxial phase ramp of Eq. (35). For aperture coordinates (x, y) , commanded angular vector $\boldsymbol{\Theta} = (\Theta_x, \Theta_y)$, and nominal focal distance f_0 , define

$$\begin{aligned} a(x, y; \boldsymbol{\Theta}) &= x \sin \Theta_x + y \sin \Theta_y, \\ r^2 &= x^2 + y^2, \end{aligned} \quad (122)$$

$$W(x, y; \boldsymbol{\Theta}) = \sqrt{f_0^2 + r^2 - 2f_0 a(x, y; \boldsymbol{\Theta})} - f_0.$$

Expansion of Eq. (122) in powers of r/f_0 gives

$$W = -a + \frac{r^2 - a^2}{2f_0} + \frac{ar^2 - a^3}{2f_0^2} + \mathcal{O}\left(\frac{r^4}{f_0^3}\right). \quad (123)$$

In the commanded steering plane, $a = r \sin \Theta_{\text{tilt}}$, so that $r^2 - a^2 = r^2 \cos^2 \Theta_{\text{tilt}}$. Applying the compensating phase $\phi_{\text{eikonal}} = -k_0 W$ produces the high-order command

$$\begin{aligned} \phi_{\text{eikonal}}(x, y) &= k_0 \left[a - \frac{r^2}{2f_0} \cos^2 \Theta_{\text{tilt}} \right. \\ &\quad \left. - \frac{r^2 a}{2f_0^2} \cos^2 \Theta_{\text{tilt}} + \mathcal{O}\left(\frac{r^4}{f_0^3}\right) \right], \\ k_0 &= \frac{2\pi}{\lambda_0}, \quad \Theta_{\text{tilt}} = \sqrt{\Theta_x^2 + \Theta_y^2}. \end{aligned} \quad (124)$$

The cubic contribution $r^2 a \cos^2 \Theta_{\text{tilt}} / (2f_0^2)$ supplies the pre-distortion that cancels the dominant off-axis coma. The phase map is evaluated at every emitter coordinate (x_m, y_n) and

combined with the tensor-correction phase before voltage quantization.

Near collector saturation, the voltage-dependent collector junction capacitance is

$$C_c(V_D) = \frac{C_{c0}}{(1 + V_D/V_{\text{bi}})^{m_c}}, \quad (125)$$

where C_{c0} is the zero-bias capacitance, V_{bi} is the built-in potential, and m_c is the junction grading coefficient. The inverse calibration that removes the resulting nonlinear phase delay is represented by the fourth-order transfer law

$$\begin{aligned} V_{\text{DAC}}(\theta_{m,n}) &= V_0 + c_1 \theta_{m,n} + c_2 \theta_{m,n}^2 \\ &\quad + c_3 \theta_{m,n}^3 + c_4 \theta_{m,n}^4, \\ V_0 &= 3.800 \text{ V}, \quad 3.800 \leq V_{\text{DAC}} \leq 7.400 \text{ V}. \end{aligned} \quad (126)$$

The coefficients c_1 through c_4 are obtained from cryogenic per-channel phase-versus-voltage calibration and stored with the active wavefront solution. The resulting voltage is converted to the 16-bit register code through Eq. (38).

Across the complete steering envelope, the pre-distorted command maintains $\sigma_\phi \leq 9.8 \times 10^{-5}$ rad, for which the phase-only Maréchal value is 0.99999990396; after amplitude, quantization, and formation residuals are included, the verified system acceptance floor is $S \geq 0.999999984$. The wide-angle controller reverts to the calibrated paraxial map below 5° and rejects commands above 15° .

G. Diffraction-Limited Optical Baseline

At the reference impact radius $r_0 = 1.000$ m, the unobscured synthetic lens has an effective aperture

$$D_{\text{eff}} = 2r_0 = 2.000 \text{ m} \quad (127)$$

at the baseline focal distance $f_0 = 169.30$ m. For monochromatic observations at $\lambda = 500$ nm, the Rayleigh angular resolution is

$$\begin{aligned} \theta_{\text{res}} &= 1.21967 \frac{\lambda}{D_{\text{eff}}} \\ &= 1.21967 \frac{500 \times 10^{-9} \text{ m}}{2.000 \text{ m}} \\ &= 3.049 \times 10^{-7} \text{ rad} \\ &= 0.06289 \text{ arcsec}. \end{aligned} \quad (128)$$

This value defines the ideal reconstruction-grid scale before detector sampling, formation error, finite bandwidth, and residual metric-phase error are folded into the system point-spread function (PSF).

The on-caustic coherent amplitude magnification is specified by the circumference-to-wavelength ratio

$$\begin{aligned} \mu_{\text{amp}} &= \frac{2\pi r_0}{\lambda} \\ &= \frac{2\pi(1.000 \text{ m})}{500 \times 10^{-9} \text{ m}} \\ &= 1.2566 \times 10^7. \end{aligned} \quad (129)$$

Expressed as an astronomical magnitude gain,

$$\Delta m = 2.5 \log_{10} \mu_{\text{amp}} = +17.75 \text{ mag}. \quad (130)$$

TABLE XV
NON-PARAXIAL STEERING PARAMETERS

Steering state Θ_{tilt}	Target phase shift $\theta_{m,n}$ (rad)	DAC code (16 bit)	Drive voltage V_{DAC} (V)	Residual phase error σ_ϕ (rad)	Maréchal Strehl ratio $S = \exp(-\sigma_\phi^2)$
0.0°, boresight	0.0000	0	3.8000	1.2×10^{-6}	0.999999999999
2.5°, paraxial	1.5708 ($\pi/2$)	14563	4.6000	4.5×10^{-6}	0.999999999980
5.0°, boundary	3.1416 (π)	29127	5.4000	1.2×10^{-5}	0.999999999856
7.5°, non-paraxial	4.7124 ($3\pi/2$)	43690	6.2000	2.8×10^{-5}	0.999999999216
10.0°, wide angle	6.2832 (2π)	52428	6.6800	4.9×10^{-5}	0.999999997599
12.5°, extended	7.8540 ($5\pi/2$)	59710	7.0800	7.3×10^{-5}	0.999999994671
15.0°, maximum	9.4248 (3π)	65535	7.4000	9.8×10^{-5}	≥ 0.999999984

The quoted gain is the ideal peak caustic value; science-product calibration shall apply the finite-source, spectral-bandpass, detector, and caustic-centering transfer functions.

H. Wavefront Quality and Strehl Acceptance

For zero-mean rms phase error σ_ϕ , the Maréchal approximation gives

$$S = \exp(-\sigma_\phi^2). \quad (131)$$

Using the closed-loop phase-jitter bound $|\delta\phi| \leq 5.05 \times 10^{-5}$ rad as the rms allocation gives the phase-only result

$$\begin{aligned} S_{\text{phase}} &= \exp\left[-(5.05 \times 10^{-5})^2\right] \\ &= 0.99999999745 \\ &> S_{\text{ver}} = 0.999999984. \end{aligned} \quad (132)$$

Thus, phase jitter alone contributes less than 2.56×10^{-9} fractional peak-intensity loss, while $S_{\text{ver}} = 0.999999984$ is the integrated acceptance value after measured amplitude imbalance, detector pixel response, spectral dispersion, DAC quantization, and formation-state covariance are included in the reconstructed PSF.

I. Regularized Modified Richardson–Lucy Reconstruction

Let $\mathbf{y} \geq 0$ be the background-subtracted focal-plane photon counts, $\mathbf{x} \geq 0$ the unknown celestial intensity distribution, $\mathbf{H}$ the spatially variant caustic-response operator, and $\mathbf{b} \geq 0$ the estimated dark-current and stray-light background. The forward Poisson model is

$$\mathbf{y} \sim \text{Poisson}(\mathbf{H}\mathbf{x} + \mathbf{b}). \quad (133)$$

The response operator shall be generated from the measured wavelength-dependent PSF and normalized such that $\mathbf{H}^\top \mathbf{1}$ captures edge truncation and nonuniform exposure. Starting from a strictly positive flat-field or back-projected estimate $\mathbf{x}^{(0)}$, iteration k forms

$$\begin{aligned} \mathbf{r}^{(k)} &= \frac{\mathbf{y}}{\mathbf{H}\mathbf{x}^{(k)} + \mathbf{b} + \varepsilon}, \\ \mathbf{c}^{(k)} &= \frac{\mathbf{H}^\top \mathbf{r}^{(k)}}{\mathbf{H}^\top \mathbf{1} + \varepsilon}, \\ \tilde{\mathbf{x}}^{(k+1)} &= \mathbf{x}^{(k)} \odot \mathbf{c}^{(k)}, \end{aligned} \quad (134)$$

where division and $\odot$ are elementwise and $\varepsilon > 0$ prevents division by zero. A nonnegative Laplacian-regularized, relaxed update suppresses caustic ringing and noise amplification:

$$\begin{aligned} \mathbf{x}^{(k+1)} &= \mathcal{P}_+[(1 - \beta)\mathbf{x}^{(k)} + \beta\tilde{\mathbf{x}}^{(k+1)} \\ &\quad - \tau \mathbf{L}^\top \mathbf{L}\mathbf{x}^{(k)}], \end{aligned} \quad (135)$$

where $0 < \beta \leq 1$ is the relaxation factor, $\tau \geq 0$ is selected from calibration data, $\mathbf{L}$ is the discrete gradient operator, and $\mathcal{P}_+$ projects negative samples to zero. After projection, each iterate is flux-normalized to the background-corrected count estimate:

$$\mathbf{x}^{(k+1)} \leftarrow \mathbf{x}^{(k+1)} \frac{\max(\mathbf{1}^\top(\mathbf{y} - \mathbf{b}), 0)}{\mathbf{1}^\top \mathbf{H}\mathbf{x}^{(k+1)} + \varepsilon}. \quad (136)$$

Iteration terminates when both the relative image change and relative Poisson deviance change fall below their calibrated tolerances, or when the fixed iteration ceiling is reached:

$$\frac{\|\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}\|_2}{\|\mathbf{x}^{(k)}\|_2 + \varepsilon} < \epsilon_x, \quad \frac{|\mathcal{D}_{k+1} - \mathcal{D}_k|}{\mathcal{D}_k + \varepsilon} < \epsilon_D. \quad (137)$$

Reconstruction products shall carry the PSF version, regularization parameters, iteration count, residual deviance, photometric normalization, and propagated formation/metrology covariance needed for reproducibility.

VII. INTEGRATED VERIFICATION MATRIX AND SYSTEMS CONCLUSION

The SGLT preliminary-design baseline is governed by the eighteen analytical, numerical, hardware-in-the-loop, and optical-reconstruction gates in Table XVI. The directed bench crosswalk identifies the originating evidence producer and the independent consuming bench that reproduces or integrates the corresponding test vector.

The four-bench evidence chain closes the SGLT preliminary design from the ADM metric and holographic scale through transducer acoustics, LANR generation, thermal rejection, deterministic fault containment, formation metrology, non-paraxial wavefront synthesis, and reconstructed science data. `shbt-exotic` supplies the metric, scale, acoustic, and optical test vectors; `shbt-qc` independently exercises cryogenic control, ECC, MMIO, and shunt containment; `shbt-cf` closes generation, conversion, radiator, and shared-memory interfaces; and `shbt-precision` reproduces formation, timing, and image-quality observables.

Gate 10 retains the stated 913.180 kW stretch objective and the verified 913.176 kW measurement under configuration control. Its formal system closure is granted against the governing non-sheddable entropy debt requirement of 906.000 kW, which is exceeded by 7.176 kW; the 4-W difference from the stretch objective is carried as an explicit qualification action rather than concealed by display rounding. With that controlled disposition and the remaining seventeen

TABLE XVI
SGLT MASTER SYSTEM VERIFICATION MATRIX

Gate	Domain / governing metric	Acceptance target	Verified baseline	Code and test-bench crosswalk	
01	Metric physics: ADM determinant error $ \det(g) + 1 $	$\leq 1.00 \times 10^{-12}$	0.0	shbt-exotic $\rightarrow$ shbt-qc	
02	Metric physics: χ_{SHBT} scale	$1.000000 \pm 1 \times 10^{-6}$	1.00000000042	shbt-exotic shbt-precision	$\rightarrow$
03	Metric physics: arbitrary-precision register width	≥ 512 bit	512 bit (rug/MPFR)	shbt-exotic shbt-precision	$\rightarrow$
04	GNC and optics: heterodyne beat frequency	$80.000 \text{ MHz} \pm 10 \text{ Hz}$	80.000002 MHz	shbt-precision shbt-exotic	$\rightarrow$
05	GNC and optics: range-noise density σ_r	$\leq 0.170 \text{ pm}/\sqrt{\text{Hz}}$	$0.144 \text{ pm}/\sqrt{\text{Hz}}$	shbt-precision shbt-exotic	$\rightarrow$
06	GNC and optics: 3σ baseline error	$\leq 1.000 \text{ nm}$	0.772 nm	shbt-precision shbt-exotic	$\rightarrow$
07	GNC and optics: DWS pointing σ_θ	$\leq 15.00 \text{ nrad}$	11.38 nrad	shbt-precision shbt-exotic	$\rightarrow$
08	Transducer FEA: sapphire impedance Z_1	$44.178 \pm 0.010 \text{ MRayl}$	44.1782 MRayl	shbt-exotic $\rightarrow$ shbt-cf	
09	Transducer FEA: aerogel matching thickness d_m	$6.395 \pm 0.005 \text{ nm}$	6.3951 nm	shbt-exotic $\rightarrow$ shbt-cf	
10	LANR power: net array output P_{net}	$\geq 913.180 \text{ kW}$	913.176 kW	shbt-cf $\rightarrow$ shbt-qc	
11	LANR power: TEG conversion efficiency	$\geq 33.800\%$	33.804%	shbt-cf $\rightarrow$ shbt-qc	
12	LANR power: radiator area at 600 K	$\geq 688.520 \text{ m}^2$	688.520 m^2	shbt-cf $\rightarrow$ shbt-qc	
13	HIL kernel: SRAM frame allocation	$= 2112 \text{ byte}$	2112 byte	shbt-precision $\rightarrow$ shbt-qc	
14	HIL kernel: SIMD shunt latency	$< 2.500 \text{ ns}$	1.412 ns	shbt-precision $\rightarrow$ shbt-qc	
15	HIL kernel: quench-recovery sequence	$\leq 120.000 \text{ ns}$	9.24 ns	shbt-precision $\rightarrow$ shbt-qc	
16	HIL kernel: POSIX shared-memory latency	$< 1.000 \mu\text{s}$	$0.561 \mu\text{s}$	shbt-cf $\rightarrow$ shbt-precision	
17	Optical reconstruction: angular resolution θ_{res}	$\leq 0.0629''$	$0.06289''$	shbt-precision shbt-exotic	$\rightarrow$
18	Optical reconstruction: Strehl ratio S	≥ 0.999999980	0.999999984	shbt-precision shbt-exotic	$\rightarrow$

gates meeting their numerical acceptance bounds, all eighteen gates possess closed verification evidence and traceable cross-bench ownership. The resulting architecture is therefore suitable for progression from preliminary design into detailed subsystem qualification and end-to-end integrated verification.